JAMA Psychiatry | Original Investigation

Adjuvant Psychotherapies to Prevent Relapse in Bipolar Disorder
A Randomized Clinical Trial

Martin Hautzinger, PhD; and the A2 BipoLife Consortium

IMPORTANCE Several psychotherapy protocols have been evaluated as adjuncts to pharmacotherapy for patients with bipolar disorder (BD). Little is known about their comparative effectiveness.

OBJECTIVE To compare the effectiveness of 2 types of group psychotherapy, skill-oriented, material-based cognitive behavioral therapy (SEKT) and supportive, patient-centered, emotion-focused therapy (FEST), to prevent relapse in patients with euthymic BD.

DESIGN, SETTING, AND PARTICIPANTS This was a large, observer-blind, randomized clinical trial conducted over 18 months (posttreatment after 6 months; follow-up at 12 and 18 months). In addition to psychiatric care as usual (including mood-stabilizing medication), each participant at 9 clinical outpatient units in Germany received 24 hours of group psychotherapy over 4, full-day sessions spread over 5 months. Patients with euthymic BD type 1 (BD 1) or BD type 2 (BD 2) between the ages of 18 and 50 years were randomly assigned to 1 of 2 forms of psychotherapy, SEKT or FEST. Independent clinicians blinded to patient grouping performed assessments using structured interviews (Structured Clinical Interview for *DSM* Disorders and Longitudinal Interval Follow-Up Evaluation) and self-rating and clinician rating for inclusion criteria and outcome. Kaplan-Meier survival curves were calculated for time to relapse. Cox proportional hazards statistics and propensity score matching were calculated for the multivariate analysis. Study data were analyzed from March 2020 to September 2022.

INTERVENTIONS SEKT intervention is a structured cognitive behavioral therapy integrating elements of interpersonal social rhythm therapy, and of mindfulness-based cognitive therapy. FEST psychotherapy has its roots in emotion-focused, supportive, and nondirective therapy.

MAIN OUTCOMES AND MEASURES Recurrence of a new affective episode assessed by blinded interviewer with the LIFE interview. In addition, self-rating and clinician rating of depressive and mania symptoms as well as level of social functioning were assessed.

RESULTS Of 348 screened referrals, 305 patients (median [IQR] age, 34 [18-50] years; 162 male [53%]) with euthymic BD 1 or BD 2 were included in the study. A total of 207 patients (68%) had BD 1, 98 (32%) had BD 2, and 278 (91%) received psychiatric care. Both therapies were equally effective in preventing recurrence of a new episode. Outcome (higher rate of new episodes) was not predicted by kind of treatment (SEKT: 69 [49%] relapse; FEST: 63 [46%] relapse) but was predicted by BD 2, comorbidity, attending all sessions, and the interaction of type of treatment by BD 1 or 2. Patients with BD 2 had the highest rate of relapse (60 [61%] relapse), in particular, when treated by SEKT (39 [70%] relapse).

CONCLUSIONS AND RELEVANCE Results of this randomized clinical trial revealed that a structured, skill-oriented, material-based cognitive behavioral therapy (SEKT) and a supportive, patient-centered, emotion-focused therapy (FEST) were equally effective in preventing relapse of affective episodes when delivered in a new, intensive group format. Additionally, there were baseline factors, in particular BD 2, that influenced outcomes.

TRIAL REGISTRATION ClinicalTrials.gov Identifier: NCT02506322

Visual Abstract

Supplemental content

Author Affiliations: Department of Psychology, Clinical Psychology and Psychotherapy, Eberhard Karls University Tuebingen, Tuebingen, Germany.

Group Information: The A2 BipoLife Consortium members appear at the end of this article.

Corresponding Author: Martin Hautzinger, PhD, Department of Psychology, Clinical Psychology and Psychotherapy, Eberhard Karls University, Schleichstrasse 4, 72076 Tuebingen, Germany (hautzinger@uni-tuebingen.de).

JAMA Psychiatry. 2024;81(9):855-862. doi:10.1001/jamapsychiatry.2024.1310
Published online June 5, 2024.

© 2024 American Medical Association. All rights reserved.

Bipolar disorder (BD) is a severe mental disease with a lifelong course and considerable morbidity and mortality. Cohort studies demonstrate that approximately 60% of patients with BD recover from an episode, and one-half of them experience recurrence within 2 years despite mood-stabilizing medication.[1]

A growing body of literature suggests that combining pharmacotherapy and psychotherapy is more effective in treating patients with BD than pharmacotherapy alone. As an adjuvant therapy, psychotherapy helps patients with BD in improving their medication compliance, awareness of illness management strategies, and ability to cope with stress, which may result in better outcomes over periods of 1 to 2 years.[2,3] Among the psychological therapies that are potential adjuncts to medications for patients with BD, cognitive-behavioral therapy (CBT) is a promising treatment option but has inconclusive findings.[4] Randomized clinical trials (RCTs) have shown benefits of CBT as an adjunct to mood stabilizers for preventing relapse, relieving symptoms, and enhancing drug adherence. Several meta-analyses have evaluated the efficacy of CBT for BD.[5-7] These studies have demonstrated that CBT has a small impact on mood symptoms, but the evidence remains inconclusive due to limited data, and some trials have found no effects overall.[8,9]

Several psychotherapy protocols have been evaluated as adjuncts to pharmacotherapy for patients with BD, but little is known about their comparative effectiveness. Miklowitz et al[10] identified 20 RCTs including 3863 participants with BD that compared pharmacotherapy plus manualized psychotherapy (CBT, family or conjoint therapy, interpersonal therapy, or psychoeducational therapy) with pharmacotherapy plus treatment as usual as the control. Psychotherapies were associated with lower recurrence rates than these nonactive control treatments (odds ratio, 0.56; 95% CI, 0.43-0.74). Already more than a decade ago, we[11] published an RCT including only 76 patients with BD. We found that an active, supportive, patient-centered therapy (SPT) was equally effective in reducing relapse rates than CBT.

To overcome some limitations of earlier studies and to cross-evaluate our previous findings, we conducted a large multicenter study comparing 2 types of active psychotherapy adjuvant to psychiatric care plus mood-stabilizing medication (eg, lithium) to prevent recurrence of a new affective episode in patients with euthymic BD over 18 months. We investigated several moderating variables (eg, age, number of previous episodes, comorbidity, etc) and introduced a new format for group therapy.

Key Points

Question Do skill-oriented, material-based cognitive behavioral therapy (SEKT) and supportive, patient-centered, emotion-focused therapy (FEST) forms of psychotherapy prevent relapse in patients with euthymic bipolar disorder?

Findings In this randomized clinical trial including 305 patients with bipolar disorder, both SEKT and FEST psychotherapies were equally effective in preventing recurrence of a new episode. Outcome (higher rate of new episodes) was not predicted by type of treatment but was predicted by bipolar 2 disorder, comorbidity, attending all sessions, and the interaction of type of treatment by bipolar disorder 1 or 2.

Meaning The provision of empathy, verbalizing emotions, and support are sufficient ingredients in relapse prevention; patients with bipolar 2 disorder need more specific treatment adapted to their course of illness.

Methods

This study was conducted from September 2016 to March 2021 in 9 urban academic clinical centers in Germany. The study protocol has been previously published (Supplement 1).[12] All study procedures were approved by the institutional review boards and ethic committees at each study site. Study participants provided informed consent after receiving a complete study description. This study followed the Consolidated Standards of Reporting Trials (CONSORT) reporting guidelines.

Study Design

This was a 20-week (5-month), evaluator-blind, parallel-group, RCT in adult outpatients with BD recruited from psychiatrist referrals, the outpatient units of study sites, and advertisements (flyer, online, informational letter) to local psychiatric practices. Participants, all receiving psychiatric treatment as usual (standardized medication, visits every 3 months, and laboratory controls), were randomly assigned either to skill-oriented, material-based CBT (SEKT) or to supportive, emotion-focused SPT (FEST) and followed up for 1 year after treatment.

Participants and Inclusion and Exclusion Criteria

Participants had to be between 18 and 50 years of age, meeting *DSM-5* criteria for BD type 1 (BD 1) or BD type 2 (BD 2), in a remitted (euthymic) state for at least 4 weeks, confirmed by Structured Clinical Interview for *DSM* Disorders I (SCID-I), a score of less than 3 on the Longitudinal Interval Follow-Up Evaluation (LIFE) Psychiatric Status Rating (PSR) scale, and a score of 10 or less on the Quick Inventory of Depressive Symptoms–clinician rated (QIDS-C) scale and 12 or less on the Young Mania Rating Scale (YMRS). All had to be in stable psychiatric care and taking mood stabilizing medication at intake and through the study protocol. The rationale behind these inclusion criteria was the study focus on relapse prevention and time to recurrence of a new affective episode. Patients with BD at study entry had to be in a symptom-free state but receiving psychiatric care and taking mood-stabilizing medication. All patients had European ancestry.

Exclusion criteria were psychotic or mental disorder, current alcohol or drug dependence, acute suicidal or homicidal ideation or requiring higher level of psychiatric care, borderline or antisocial personality disorder, current participation in psychotherapy, last illness episode more than 2 years prior, not fluent in the German language, not taking medication (mood stabilizer), or not giving informed consent.

© 2024 American Medical Association. All rights reserved.

Sample Size Calculation and Characteristics

Sample size calculation was based on a 2-sided log-rank test under the following specifications: (1) study duration of 48 months, with patient recruitment during the first 36 months and (2) proportion of dropouts in the control group and in the treatment group of 12% each (based on experiences in former studies[11]). Under the a priori-defined condition that the proportion of relapse-free patients should improve from 50% in the control group to 70% in the treatment group during the first year of treatment, this required a sample size of 300 (n = 150 per treatment arm) for statistical significance ($1 - \beta = 80\%$; $\alpha = .05$).

Measures

Evaluators blind to treatment assignment conducted assessments at baseline, at posttreatment (2-4 weeks after the last therapy session), and at 6 and 12 months after posttreatment assessment (follow-up visits). Diagnoses were confirmed with the SCID-I and SCID-II. Depressive, manic, or hypomanic episodes were assessed every 6 months by specially trained, independent clinicians (evaluators) performing the LIFE interview.[13] The LIFE PSR scale (1-6 per week) was the major outcome measure. A rating of 5 and 6 for 2 weeks in a row was considered a new illness episode. Time in a new episode was calculated as well as total time with severe symptoms. The LIFE interview was performed every 6 months by independent clinicians (blind to the intervention) after special training (24 hours of workshop) in this interview and the PSR rating. Additionally, all evaluators had to conduct at least 2 LIFE interviews with patients who were not study participants. These training interviews were videotaped and evaluated by supervisors with several years of LIFE interview experience in other research projects.

We used the QIDS-self-rated (QIDS-SR)[14] and QIDS-C[14] instruments to measure depressive symptoms, and the YMRS[15] (clinician rated) and the Altman Self-Rating Mania Scale (ASRM)[16] to measure manic symptomatology. The area of social functioning in various living areas was evaluated by the Global Assessment of Functioning (GAF) and the Functional Assessment Short Test (FAST)[17] ratings.

Allocation and Randomization

Randomization and allocation sequence was generated by an independent biostatistician who was employed at a biostatistics company (Numerous Ltd, London, UK) and who was not otherwise involved in any study procedures.

A center-specific block randomization was performed in SAS, version 9.4 (SAS Institute) based on uniformly distributed random numbers, with a block length of 6 for 16 blocks per center (yielding 96 potential group assignments per center). The center-specific randomization lists were available exclusively for the principal investigator (M.H.) who informed study centers of a patient's group assignment when a center made the request to enroll a new patient. The randomization protocol ensured that each study center conducted both psychotherapies. We did not plan to stratify for BD 1 or BD 2 due to our large study sample.

Interventions and Adherence to Protocol

All participants were evaluated by a psychiatrist of their choice. These clinicians had to follow the basic treatment rules of national treatment guidelines for BD. This implies providing medication (mood stabilizer) and regular, short appointments for their patients to check blood serum, compliance with treatment, and to give support and advice for medical problems. The personal psychiatrist was also responsible to adjust medication (dosage change, new medication) with symptom changes, to check for suicidality, or to admit the patient to a hospital, if necessary.

In addition to usual care, each participant received 24 hours of group psychotherapy by the same therapists in a new group format. Participants were evaluated 4 times for a full day (from 9 AM to 5 PM) in an open group setting of 5 to 7 participants, always with 1 study therapist and an assistant. Each treatment day was split into 4 blocks, each approximately 90 minutes long with breaks of 30 to 45 minutes between each group session. The treatment days were 4 to 5 weeks apart, resulting in a maximum of 24 hours of psychotherapy over 5 months.

There were 2 forms of psychotherapy delivered to the study participants, depending on random assignment. Our SEKT intervention is a structured CBT[18] integrating elements of interpersonal social rhythm therapy, of mindfulness-based cognitive therapy, and of meta-cognitive therapy (eAppendix 8 in Supplement 2). The rationale of SEKT is to teach participants certain skills to overcome symptoms and to prevent recurrence of new episodes. There is a SEKT treatment manual available including all material and exercises for group sessions.

Our FEST psychotherapy has its roots in supportive, emotion-focused, and client-centered therapy.[19] There is no organized, well-planned structure for a treatment day. The rationale of FEST is to help participants understand their feelings and emotional processes in personal problems and to support each other and discover their personal resources to prevent recurrence of new episodes. There is a FEST treatment manual available containing information on the types of interactions therapists are permitted (and not permitted) to have with the client groups (eAppendix 8 in Supplement 2).

Study therapists (master's- or doctoral-level psychologists or psychiatrists) all had several years of clinical experience in the application of CBT. All study therapists underwent additional training workshops (24 hours) in both study treatments (SEKT and FEST). In addition, before being accepted as a study therapist, they had to demonstrate their skills in leading groups with SEKT and FEST. All group sessions were videotaped to monitor fidelity and adherence to treatment protocol. At all sites, all therapists participated in monthly supervision with local experts, and all met once a year for 2 days with national experts.

Adherence to the treatment protocols was assessed after termination of all study therapies at all sites using a 24-item rating scale containing 10 items covering SEKT and 10 items covering FEST interventions and 4 general, nonspecific items. Blind to selection of videotapes, all study therapists were evaluated by 3 trained raters who were not involved in the study and were uninformed about the kind of treatment on the videotapes, the origin of the videotapes, and the identity of the thera-

© 2024 American Medical Association. All rights reserved.

Figure 1. Consolidated Standards of Reporting Trials Diagram

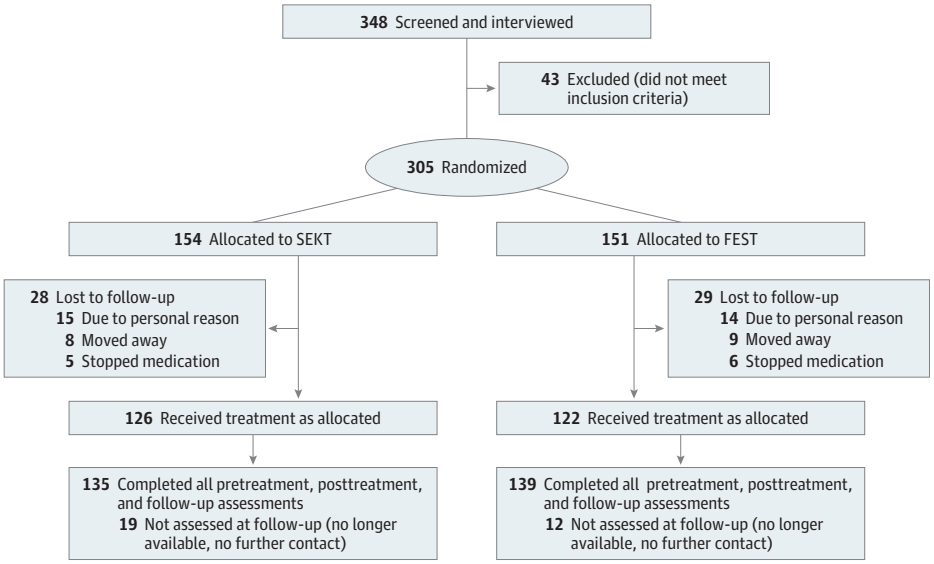


Diagram of the intention-to-treat (N = 305) and completer (N = 274) sample. FEST indicates supportive, patient-centered, emotion-focused therapy; SEKT, skill-oriented, material-based cognitive behavioral therapy.

pists. Results of adherence to protocol ratings are summarized in eAppendix 1 in Supplement 2. This protocol evaluation shows a clear difference between FEST and SEKT as well as a high adherence to each therapy manual.

Statistical Analyses

The major outcome criterion was time to recurrence of a new episode (PSR ratings, either depression or mania [or hypomania]). Kaplan-Meier survival curves were calculated for time to relapse (PSR ≥5 for 1 week for mania, 2 weeks for depression). In addition, Cox proportional hazards statistics were calculated for multivariate analyses of survival curves. The proportional hazard condition states that covariates are multiplicatively related to the hazard (relapse). Furthermore, propensity score matching (PSM)[20] was used post hoc to estimate the effect of the interventions by matching individuals in 1 group with individuals in the other group for covariates that predict receiving the treatment. This results in 2 groups (SEKT, FEST) of similar characteristics to estimate the impact of the intervention. PSM is indicated in observational studies and in RCTs where variable balance failed. The following variables were used to match to groups: kind of psychotherapy, BD 1 or BD 2, comorbidity, compliance, age, sex, previous affective episodes, treatment with lithium, history of psychotherapy, and study site. Data were analyzed from March 2020 to September 2022 using JASP version 0.18.1 (open source, University of Amsterdam); SAS, version 9.4 (SAS Institute); and R, version 4.0.0 (R Foundation). All *P* values were 2-sided, and $P < .05$ was considered statistically significant.

Results

As the CONSORT diagram shows (**Figure 1**), 348 individuals were screened to yield 305 persons (median [IQR] age, 34 [18-50] years; 143 female [47%]; 162 male [53%]) eligible for randomization. A total of 207 patients (68%) had BD 1, 98 (32%) had BD 2, and 278 (91%) received psychiatric care. During the intervention phase, 57 of 305 randomized participants were lost to follow-up (dropout rate, 18.7%), but some of them were still willing to be evaluated at follow-up. In total, we assessed 274 participants and lost 31 (10%). There was no difference in dropout rate between the 2 study interventions. **Table 1** summarizes the baseline characteristics of all 305 study participants. There were no statistically significant differences between the SEKT and FEST groups regarding baseline characteristics, except that there were more patients with BD 2 randomized to the SEKT group (BD 1 = 99 [48%]; BD 2 = 55 [56%]) than to the FEST group (BD 1 = 108 [52%]; BD 2 = 43 [44%]; $\chi^2 = 17.98$; $P = .01$).

In a first step, we analyzed the self-rating and clinician rating (QIDS, YMRS, ASRM, GAF) for both interventions across all assessment points (**Table 2**).

The assessment of social functioning (GAF scores) across total study period can be found in eAppendix 2 in Supplement 2. Overall, our study sample had euthymic BD without active symptoms, and all parameters were in the normal range. At all 3 points in time across participants, we found no sign of depression (QIDS-SR/QIDS-C; PSR-depression), mania (ASRM, YMRS, PSR-mania), and medium- to high-level of social functioning (FAST, GAF). Neither treatment, time, nor their interaction were statistically significant.

Use of LIFE and PSR ratings week by week through treatment and follow-up lead to a picture as shown in **Figure 2**. New affective episodes (manic or depressive episodes based on *DSM-5* criteria) occurred throughout the study period and in both treatments. Unexpectedly, participants in the FEST group had a much more positive survival curve and outcome than those in the SEKT group (FEST, 88 of 142 [62%] without relapse; SEKT, 68 of 138 [49%] without relapse). Both a log-

© 2024 American Medical Association. All rights reserved.

Table 1. Baseline Characteristics of Study Sample (N = 305)[a]

Groups	SEKT	FEST	Total
Sample, No. (%)	154 (50.5)	151 (49.5)	305
Age, median (IQR), y	35 (18-50)	33 (18-49)	34 (18-50)
Sex, No. (%)			
Female	74 (51)	69 (49)	143 (47)
Male	80 (49)	82 (51)	162 (53)
Diagnosis, No. (%)			
BD 1	99 (48)	108 (52)	207 (68)
BD 2	55 (56)	43 (44)	98 (32)[b]
Previous episodes, median	7	7	7
Comorbidity (additional psychopathology), No. (%)	53 (35)	45 (28)	98 (32)
School education, No. (%), y			
12	85 (55)	95 (62)	180 (59)
10-11	55 (37)	51 (33)	106 (35)
9	10 (7)	9 (5)	19 (6)
Professional education, No. (%)			
Yes	135 (89)	130 (85)	265 (87)
No	16 (11)	24 (15)	40 (13)
Employment status, No. (%)			
Unemployed, on pension	55 (38)	52 (32)	107 (35)
Employed (part time, full time)	77 (51)	76 (50)	153 (50)
Professional on-the-job training	21 (12)	25 (18)	46 (15)
Sick leave during last 5 y, median (IQR), mo	8 (4-35)	7 (3-37)	8 (3-37)
Income per year, No. (%), €[c]			
>36 000	74 (25)	70 (22)	72 (24)
24 000-35 000	44 (13)	46 (16)	45 (15)
12 000-23 000	44 (30)	38 (25)	82 (27)
<12 000	52 (32)	54 (37)	106 (35)
Psychiatric care, includes taking medication, No. (%)	142 (93)	136 (89)	278 (91)
Mood stabilizer: lithium medication, No. (%)	81 (53)	84 (56)	165 (54)
Other medications: antidepressant, No. (%)	59 (38)	60 (39)	119 (39)
Other medication: neuroleptics, No. (%)	62 (40)	63 (42)	125 (41)
Other, nonspecified medication, No. (%)	39 (14)	40 (16)	79 (15)
Previous inpatient treatment, No. (%)	131 (87)	128 (84)	259 (85)
Previous outpatient psychotherapy, No. (%)	107 (72)	103 (66)	210 (69)

Abbreviations: BD, bipolar disorder; FEST indicates supportive, patient-centered, emotion-focused therapy; SEKT, skill-oriented, material-based cognitive behavioral therapy.

[a] There were no statistically relevant differences in any baseline characteristics between SEKT and FEST study arm except BD 1 and BD 2.

[b] $\chi^2 = 17.98$; $P \leq .01$.

[c] Conversion from euro to US dollar: divide by 1.07123.

rank test ($\chi^2 = 4.86$; $P \leq .02$) and a Wilcoxon test ($\chi^2 = 4.62$; $P \leq .03$) resulted in a significant difference between the 2 survival curves with a new affective episode as the survival criterion. At the end of group therapy, there was no statistically significant difference between treatment groups, but 22% of patients (30 of 138) in the SEKT group and 19% of patients (27 of 142) in the FEST group had experienced either a depressive or manic (or hypomanic) episode (PSR ≥5). Mean (SD) duration of a recurrent, new affective episode during the study period was not different for the 2 psychotherapy groups (SEKT, 5.98 [7.32] weeks and FEST, 5.64 [6.28] weeks).

eAppendixes 3 and 4 in Supplement 2 show the differences between participants with BD 1 and BD 2. Participants with BD 2 had many more relapses during the 18-month study period (60 [61%]), in particular, when treated by SEKT (39 [70%] relapse; log-rank $\chi^2 = 5.93$; $P \leq .02$; Wilcoxon test $P \leq .02$). Calculating multivariate analyses instead of univariate survival analysis, the factor therapy was not statistically significant (eAppendix 5 in Supplement 2). Proportional hazard modeling included the factors therapy (SEKT and FEST), BD 1 and BD 2, age, sex, comorbidity (psychopathology), number of previous episodes, previous psychotherapy, compliance with group treatment, and their interactions. eAppendixes 5 and 7 in Supplements 2 show that BD 1 and BD 2, the interaction of BD 1 and BD 2 with study therapy and with previous psychotherapy, the interaction of study therapy with comorbidity, and the interaction of sex with compliance were the only significant variables associated with outcome (recurrence of affective episode). A total of 70% of patients (32 of 46) with BD 2 had relapses when treated with SEKT (log-rank $\chi^2 = 10.46$; $P \leq .02$; Wilcoxon $\chi^2 = 10.13$; $P \leq .02$) compared with FEST (51% of patients [19 of 37] had relapses). In addition, comorbidity interacted with psychotherapy (log-rank $\chi^2 = 14.25$; $P \leq .003$; Wilcoxon $\chi^2 = 13.65$; $P \leq .003$) leading to reduced prevention effects in both forms of treatment. Patients with BD and comorbid psychopathology had more and earlier re-

© 2024 American Medical Association. All rights reserved.

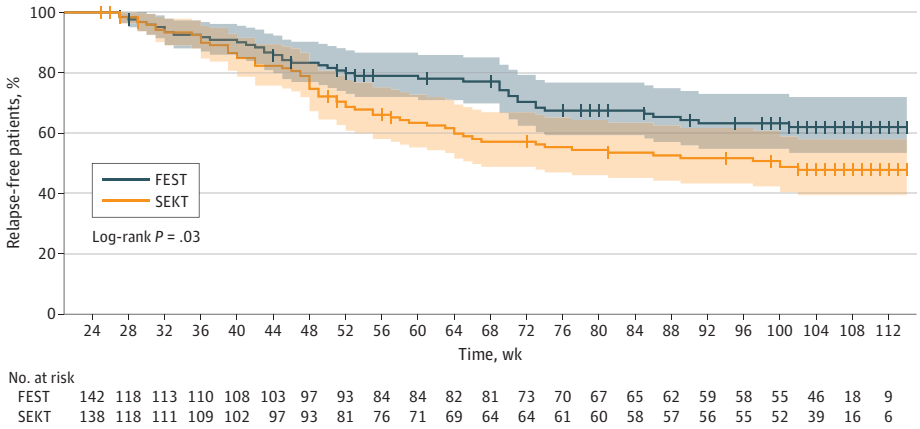
Table 2. Self and Clinician Ratings Across Assessments Points for Both Interventions (N = 305)

Rating test	Geometric mean (SE) [95% CI]			Statistics[a]
	Baseline	End of treatment	End of follow-up	
QIDS-SR	6.18 (4.52) [4.04-5.05]	6.39 (4.51) [3.95-5.13]	6.37 (4.29) [3.65-5.03]	Time: P = .63 (F = 0.58) Therapy: P = .73 (F = 0.12) Therapy × time: P = .44 (F = 0.90)
QIDS-C	5.54 (3.34) [2.98-3.73]	5.78 (3.57) [3.09-4.11]	5.72 (3.61) [3.05-4.25]	Time: P = .18 (F = 2.26) Therapy: P = .83 (F = 0.05) Therapy × time: P = .43 (F = 0.92)
ASRM	1.81 (1.06) [0.88-1.24]	1.73 (1.1) [0.90-1.32]	1.47 (0.82) [0.63-1.02]	Time: P = .20 (F = 2.13) Therapy: P = .17 (F = 1.86) Therapy × time: P = .45 (F = 0.88)
YMRS	1.98 (0.87) [0.72-1.03]	1.96 (0.95) [0.75-1.16]	1.93 (0.87) [0.66-1.11]	Time: P = .85 (F = 0.26) Therapy: P = .18 (F = 1.78) Therapy × time: P = .37 (F = 1.05)
FAST	10.20 (6.81) [6.03-7.68]	10.29 (6.20) [5.35-7.16]	10.41 (6.42) [5.41-7.59]	Time: P = .82 (F = 0.31) Therapy: P > .99 (F = 0.36) Therapy × time: P = .97 (F = 0.07)
PSR1 depression	2.25 (2.10) [1.99-2.21]	1.76 (1.67) [1.59-1.75]	1.81 (1.68) [1.57-1.79]	Time: P < .001 (F = 9.29) Therapy: P = .66 (F = 0.19) Therapy × time: P = .42 (F = 0.95)
PSR2 mania	1.40 (1.35) [1.29-1.40]	1.24 (1.20) [1.16-1.24]	1.23 (1.19) [1.15-1.24]	Time: P < .001 (F = 9.51) Therapy: P = .63 (F = 0.23) Therapy × time: P = .65 (F = 0.54)

Abbreviations: ASRM, Altman Self-Rating Mania Scale; FAST, Functional Assessment Short Test; PSR1, Psychiatric Status Rating 1; PSR2, Psychiatric Status Rating 2; QIDS-C, Quick Inventory of Depressive Symptoms–clinician rated; QIDS-SR, Quick Inventory of Depressive Symptoms–self-rated; YMRS, Young Mania Rating Scale.

[a] Statistics: Results from a repeated measures analysis of covariance using model log(Y+1) = therapy + time + therapy × time + random(patient[therapy]), whereby Y denotes the rating parameter. Log transformation has been used to reduce skewness in Y and factor patient has been nested under therapy applying a random factor to account for longitudinal assessments over time.

Figure 2. Kaplan-Meier Curves for Time to First Relapse (New Episode)



Kaplan-Meier curves for time to first relapse (new episode) with pointwise 95% confidence bands (shaded areas). At the end of follow-up, 62% and 49% of participants in the supportive, patient-centered, emotion-focused therapy (FEST) and skill-oriented, material-based cognitive behavioral therapy (SEKT) groups, respectively, were relapse free. Log-rank test χ^2_1= 4.86 (P < .03).

lapses of affective episodes, independent of the type of psychotherapy. The interaction of sex and compliance comes from male participants missing more psychotherapy sessions than female participants. Participants who missed 2 or more group sessions had a more unfavorable outcome and earlier relapses (proportional hazard effect size estimate, −0.50; 95% CI, −0.79 to −0.20; P ≤.001). As already mentioned (Table 1), unexpectedly, there were significantly more patients with BD 2 who were randomized to the SEKT group. This uneven distribution had a major impact on outcome. Patients with BD 2 did not respond as well to CBT (SEKT) but showed, in general, an unfavorable course of illness (more previous episodes, more and earlier relapses).

Using PSM (nearest neighbor matching, caliper 0.05, sample size = 174) leads to **Figure 3** showing almost identical survival curves for our 2 group therapies (log-rank χ^2 = 0.28; P ≤.60; Wilcoxon χ^2 = 0.41; P ≤.52). Outcome (higher rate of new episodes) was not predicted by kind of treatment (SEKT: 49% [42 of 87] relapse; FEST: 46% [40 of 87] relapse). Previous differences were not influenced by kind of therapy but more by confounding variables, in particular, BD 2 and comorbidity (eAppendix 6 in Supplement 2).

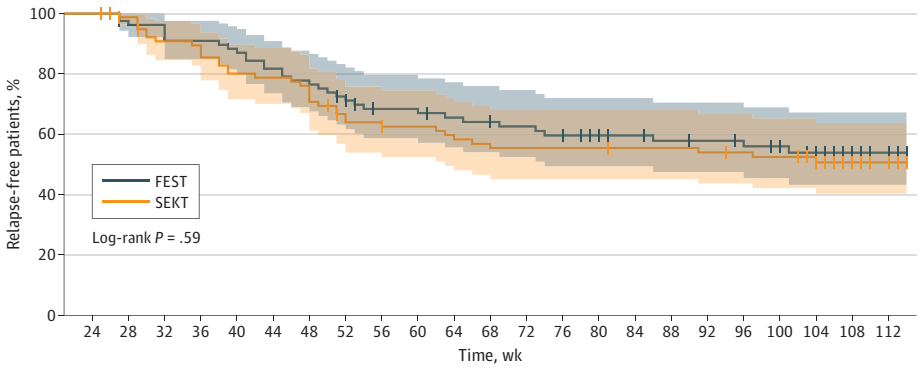
Discussion

This RCT compared 2 forms of psychotherapy adjuvant to psychiatric care and mood-stabilizing medication (mostly lithium) to prevent relapse in patients with euthymic BD 1 and BD 2 over 18 months (6 months intervention, 12 months follow-up). We expected that the well-established, evidence-based and extended CBT (SEKT) would be more efficient in preventing relapse than an unstructured, patient-centered, supportive therapy (FEST). Several reviews[2,3,5] also support this hypothesis.

However, in a previous study[11] both active, individual treatments (CBT, SPT) were similar in relapse prevention. Together with the current results, this outcome was similar to that

© 2024 American Medical Association. All rights reserved.

Figure 3. Propensity Score Matching Survival Curve



No. at risk

FEST	86	74	73	69	67	62	59	54	48	48	45	44	42	40	37	35	33	32	31	29	25	10	6
SEKT	86	75	68	67	60	59	57	48	46	44	42	40	39	39	39	38	38	37	36	35	29	11	4

Kaplan-Meier curves for time to first relapse pointwise 95% confidence bands (shaded areas). Log-rank test $\chi^2_1 = 0.28$ ($P < .60$). FEST indicates supportive, patient-centered, emotion-focused therapy; SEKT, skill-oriented, material-based cognitive behavioral therapy.

of a meta-analysis,[21,22] which found a medium effect size (Hedges $g = 0.58$) for nondirective, supportive psychotherapy in adults diagnosed with unipolar depression.

The general summary of our results is that the changes observed over time in both SEKT and FEST in subsyndromal symptoms, rates of recurrence of new episodes, and time to recurrence were similar. Outcome (higher rate of new episodes) was not predicted by kind of treatment but was predicted by BD 2, comorbidity, attendance to all therapy sessions, sex, previous psychotherapy, and the interaction of type of treatment by BD 1 or 2. Overall, patients with BD 2 had significantly more relapses (new episodes) during the study period than patients with BD 1. Patients with BD 1 benefited most by the unstructured, open, supportive FEST group treatment. Patients with BD 2 showed the least positive outcome (highest relapse rate) under SEKT (CBT).

All of our study participants had a large number of previous episodes and a majority (69%) had previous psychotherapy. For them, another structured, educational, and exercise-oriented treatment (SEKT) did not provide any new information, skills, or preventive strategies, whereas the open, supportive FEST treatment allowed for personal input, mutual support, and the activation of personal resources. This might explain why FEST was well received and successful.

Patients with BD 2 seem to be a group of patients with specific needs for psychotherapy and relapse prevention.[11,23] According to our results, future studies should either stratify for BD 1 and BD 2 or, depending on research questions, include patients with only BD 1 or BD 2. Until now, to our knowledge, there were almost no treatment outcome studies focusing on patients with BD 2. Well known from other studies,[24] compliance (attending all group sessions) and no comorbidity have a strong influence on (positive) outcome. These 2 factors should also be controlled in future studies either by excluding patients who are not compliant with treatment and those with comorbid disorders or stratifying for these variables with impact on treatment outcome. Alternatively, treatments need to be developed specifically for patients with BD and concomitant comorbidity.

Our new, full-day group therapy provided once a month was well accepted. The effects on relapse prevention were similar to those of previous studies and calculations performed in a meta-analysis.[10] We are aware that such full-day treatment sessions are a strong departure from all existing trials and from clinical practice. We see the benefit of such a treatment format in a more focused, intensive training. More patients living in rural areas can be reached and matriculated in the program, and the cost of traveling as well as air pollution are reduced. Interaction between participants during a full-day session is more intensive, and mutual support can develop while spending break time together. On the other hand, if participants experience symptoms, especially depression, a more regular schedule of sessions may be more beneficial to overcome early signs of recurrence of a new episode or to practice skills with the therapist and between therapy session. The issue of different psychotherapy formats and their effects on outcome is not well studied.

Limitations

Several factors limit our study interpretation. First of all, the number of participants with BD 1 and BD 2 was unbalanced in our study groups. We could not find any flaws in our allocation and randomization strategy. However, in future studies, it might be better to stratify for BD 1 and BD 2. Second, the group of study patients had diagnoses of BD for many years, and a large number had experienced previous episodes and psychotherapy. Newly diagnosed patients might benefit more from a structured, skill-oriented, material-based approach. All participants entered the program in a euthymic state. It remains unclear how well each treatment would perform in patients with active mood symptoms. An emotion-focused, unstructured therapy (FEST) may then even lead to an exacerbation of symptoms. Finally, the factors comorbidity and receipt of pharmacotherapy remain unclear (eAppendix 9 in Supplement 2) because the collected information was not specific enough and often incomplete.

© 2024 American Medical Association. All rights reserved.

Conclusions

In this RCT, we found evidence that a structured, skill-oriented, material-based treatment (SEKT) and a supportive, patient-centered, emotion-focused therapy (FEST) were equally effective in preventing relapse of affective episodes when delivered in an intensive group format. It seems possible that the provision of psychoeducation, empathy, and support are active and sufficient ingredients in relapse prevention for BD. The influence of BD 1 and BD 2 on treatment outcome needs further study.

ARTICLE INFORMATION

Accepted for Publication: March 27, 2024.

Published Online: June 5, 2024. doi:10.1001/jamapsychiatry.2024.1310

Author Contributions: Dr Hautzinger had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.
Concept and design: Hautzinger.
Acquisition, analysis, or interpretation of data: Hautzinger.
Drafting of the manuscript: Hautzinger.
Critical review of the manuscript for important intellectual content: Hautzinger.
Statistical analysis: Hautzinger.
Obtained funding: Hautzinger.
Administrative, technical, or material support: Hautzinger.
Supervision: Hautzinger.

Conflict of Interest Disclosures: Dr Hautzinger reported receiving grants from the German Government Minister of Education and Research during the conduct of the study.

Funding/Support: This work was supported by grant BMBF 01EE1404A-H from the German Government within the BipoLife Research Net.

Role of the Funder/Sponsor: The funder had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Group Information: Members of the A2 BipoLife Consortium were as follows: T. Stamm, MD, F. Bermpohl, MD, M. Bauer, MD, T. Kircher, MD, M. Lambert, MD, A. Fallgatter, MD, O. Gruber,MD, A. Reif, MD, T. Schulze, MD, J. Zwick, PhD, R. Niebler,PhD, K. Krisch, PhD, E. Beck, PhD, M. Wahl-Moerk, PhD, I. Lang, PhD, S. Boden, PhD, J. Fehr, PhD, J. Müller, S. Müller, S. Eck, K. Fuhr, PhD, L. Wolkenstein, PhD, I. Kunze, G. O'Malley, D. Ritter, MD, J. Conell, MD, M. Rottmann-Wolf, PhD, C. Berndt, PhD, L. Jurjanz, PhD, E. Mennigen, PhD, M. Spreer, PhD, V. Niehaus-Kraft, PhD, G. Sarikaya, MD, M. Sengutta,MD, A. Rohenkohl, F. Ruppelt, PhD, J. Fiebig, E. Quinlivian, MD, C.K. Gawron, PhD, A. Lehmann, PhD, A. Wietzke, PhD, L.M. Gassner, PhD, C. H. Attar, PhD, P. Stahl, PhD, S. Spengler, PhD, O. Shmailovich, PhD, S. Kittel-Schneider, PhD, MD, PhD, S. Biere, PhD, S. Matura, PhD, K. Hahl, PhD, B. Friedrichs, PhD, V. Oertel-Knoechel, PhD, N. Goldvach, PhD, K. Cless, PhD, J. Volkert, PhD, I. Falkenberg, PhD, MD, PhD, M. Zavorotny, PhD, I. Kluge, PhD, S. Menl, PhD, S. Betke, PhD, A. Braunisch, PhD, D. Schweiger, PhD, M. Abdel-Hamid, PhD, S. Kamping, PhD, S. Trost, PhD, J. Kalman, PhD, K. Adorjan, PhD, C. Born, PhD, F. Aldinger, PhD, H-P. Dürr, PhD.

Disclaimer: All information and materials are original and not published before.

Data Sharing Statement: See Supplement 3.

Additional Contributions: We thank David J. Miklowitz, PhD (Department of Psychiatry and Biobehavioral Sciences, Semel Institute, University of California at Los Angeles), and Thomas D. Meyer, PhD (Department of Psychiatry and Behavioral Sciences, Mood Spectrum Disorder Laboratory, University of Texas, Houston), for their comments on an earlier version of this report.

REFERENCES

1. Perlis RH, Ostacher MJ, Patel JK, et al. Predictors of recurrence in bipolar disorder: primary outcomes from the Systematic Treatment Enhancement Program for Bipolar Disorder (STEP-BD). *Am J Psychiatry*. 2006;163(2):217-224. doi:10.1176/appi.ajp.163.2.217

2. Beynon S, Soares-Weiser K, Woolacott N, Duffy S, Geddes JR. Psychosocial interventions for the prevention of relapse in bipolar disorder: systematic review of controlled trials. *Br J Psychiatry*. 2008;192(1):5-11. doi:10.1192/bjp.bp.107.037887

3. Miklowitz DJ. Adjunctive psychotherapy for bipolar disorder: state of the evidence. *Am J Psychiatry*. 2008;165(11):1408-1419. doi:10.1176/appi.ajp.2008.08040488

4. Thase ME, Kingdon D, Turkington D. The promise of cognitive behavior therapy for treatment of severe mental disorders: a review of recent developments. *World Psychiatry*. 2014;13(3):244-250. doi:10.1002/wps.20149

5. Szentagotai A, David D. The efficacy of cognitive-behavioral therapy in bipolar disorder: a quantitative meta-analysis. *J Clin Psychiatry*. 2010;71(1):66-72. doi:10.4088/JCP.08r04559yel

6. Ye BY, Jiang ZY, Li X, et al. Effectiveness of cognitive behavioral therapy in treating bipolar disorder: an updated meta-analysis with randomized controlled trials. *Psychiatry Clin Neurosci*. 2016;70(8):351-361. doi:10.1111/pcn.12399

7. Chiang KJ, Tsai JC, Liu D, Lin CH, Chiu HL, Chou KR. Efficacy of cognitive-behavioral therapy in patients with bipolar disorder: a meta-analysis of randomized controlled trials. *PLoS One*. 2017;12(5):e0176849. doi:10.1371/journal.pone.0176849

8. Scott J, Colom F, Vieta E. A meta-analysis of relapse rates with adjunctive psychological therapies compared to usual psychiatric treatment for bipolar disorders. *Int J Neuropsychopharmacol*. 2007;10(1):123-129. doi:10.1017/S1461145706006900

9. Parikh SV, Zaretsky A, Beaulieu S, et al. A randomized controlled trial of psychoeducation or cognitive-behavioral therapy in bipolar disorder: a Canadian Network for Mood and Anxiety treatments (CANMAT) study [CME]. *J Clin Psychiatry*. 2012;73(6):803-810. doi:10.4088/JCP.11m07343

10. Miklowitz DJ, Efthimiou O, Furukawa TA, et al. Adjunctive psychotherapy for bipolar disorder: a systematic review and component network meta-analysis. *JAMA Psychiatry*. 2021;78(2):141-150. doi:10.1001/jamapsychiatry.2020.2993

11. Meyer TD, Hautzinger M. Cognitive behavioral therapy and supportive therapy for bipolar disorders: relapse rates for treatment period and 2-year follow-up. *Psychol Med*. 2012;42(7):1429-1439. doi:10.1017/S0033291711002522

12. Stamm TJ, Zwick JC, O'Malley G, Sondergeld LM, Hautzinger M. Adjuvant psychotherapy in early stage bipolar disorder: study protocol for a randomized controlled trial. *Trials*. 2020;21(1):845. doi:10.1186/s13063-020-04755-8

13. Keller MB, Lavori PW, Friedman B, et al. The longitudinal interval follow-up evaluation: a comprehensive method for assessing outcome in prospective longitudinal studies. *Arch Gen Psychiatry*. 1987;44(6):540-548. doi:10.1001/archpsyc.1987.01800180050009

14. Rush AJ, Trivedi MH, Ibrahim HM, et al. The 16-Item Quick Inventory of Depressive Symptomatology (QIDS), clinician rating (QIDS-C), and self-report (QIDS-SR): a psychometric evaluation in patients with chronic major depression. *Biol Psychiatry*. 2003;54(5):573-583. doi:10.1016/S0006-3223(02)01866-8

15. Young RC, Biggs JT, Ziegler VE, Meyer DA. A rating scale for mania: reliability, validity, and sensitivity. *Br J Psychiatry*. 1978;133(5):429-435. doi:10.1192/bjp.133.5.429

16. Altman E. Rating scales for mania: is self-rating reliable? *J Affect Disord*. 1998;50(2-3):283-286. doi:10.1016/S0165-0327(98)00018-4

17. Rosa AR, Sánchez-Moreno J, Martínez-Aran A, et al. Validity and reliability of the Functioning Assessment Short Test (FAST) in bipolar disorder. *Clin Pract Epidemiol Ment Health*. 2007;3(1):5. doi:10.1186/1745-0179-3-5

18. Meyer TD, Hautzinger M. *Bipolar Disorder: A Cognitive-Behavioral Treatment Program*. 2nd ed. Beltz Psychologie Verlags Union; 2013

19. Markowitz JC. *Brief Supportive Psychotherapy: A Treatment Manual and Clinical Approach*. Oxford University Press; 2022.

20. Rosenbaum PR, Rubin D. Constructing a control group using multivariate matched sampling methods that incorporate the propensity score. *Am Stat*. 1985;39(1):33-38. doi:10.1080/00031305.1985.10479383

21. Cuijpers P, Driessen E, Hollon SD, van Oppen P, Barth J, Andersson G. The efficacy of nondirective supportive therapy for adult depression: a meta-analysis. *Clin Psychol Rev*. 2012;32(4):280-291. doi:10.1016/j.cpr.2012.01.003

22. Lam DH, Burbeck R, Wright K, Pilling S. Psychological therapies in bipolar disorder—the effect of illness history on relapse prevention: a systematic review. *Bipolar Disord*. 2009;11(5):474-482. doi:10.1111/j.1399-5618.2009.00724.x

23. Swartz HA, Rucci P, Thase ME, et al. Psychotherapy alone and combined with medication as treatments for Bipolar II Depression: a randomized controlled trial. *J Clin Psychiatry*. 2018;79(2):16m11027. doi:10.4088/JCP.16m11027

24. Trinkler C, Strauss B, Gumz A. Consideration of comorbidities in research on the effectiveness of depression treatment. Article in German. *Die Psychotherapie*. 2023;68(3):123-127. doi:10.1007/s00278-022-00634-6

© 2024 American Medical Association. All rights reserved.

EBERHARD KARLS UNIVERSITÄT TÜBINGEN

UNIVERSITÄTS KLINIKUM TÜBINGEN

Studien Operation Plan

Adjuvante Psychotherapie zur Rückfallprävention einer Bipolaren Störung – "A2BipoLife"

Gesamtleitung: Prof. Dr. Martin Hautzinger (hautzinger@uni-tuebingen.de)

Kooperationspartner:
Prof. Dr. Andreas Fallgatter,
Klinik für Psychiatrie und Psychotherapie der Universität Tübingen
Prof. Dr. Felix Bermpohl,
Klinik für Psychiatrie und Psychotherapie Charité HU Berlin
Prof. Dr. Bauer,
Klinik und Poliklinik für Psychiatrie und Psychotherapie der TU Dresden
Prof. Dr. Juckel,
Klinik für Psychiatrie und Psychotherapie der Ruhr Universität Bochum
Prof. Dr. Kircher,
Klinik für Psychiatrie und Psychotherapie der Philipps Universität Marburg a.L.
Prof. Dr. Lambert,
Klinik für Psychiatrie und Psychotherapie des UKE Hamburg
Prof. Dr. Gruber,
Klinik für Psychiatrie und Psychotherapie der Georg August Universität Göttingen
Prof. Dr. Reif,
Klinik für Psychiatrie und Psychotherapie der J.W. Goethe Universität Frankfurt a.M.
Prof. Dr. Schulze,
Klinik für Psychiatrie und Psychotherapie der Ludwigs-Maximilian Universität,
Institut für Psychiatrische Phänomik und Genomik, München
Prof. Dr. Rienhoff,
Zentrum für Medizinische Informatik, Georg August Universität Göttingen
PD Dr. Hans-Peter Dürr
Firma Numerus Ltd, Biostatistik Tübingen-London

Forschungsverbund „BipoLife" im „Forschungsnetz psychische Erkrankungen" des BMBF,
Förderkennzeichen: 01EE1404C (Bescheid vom 1.12.2014)
Registrierung Clinical Trials: **NCT02506322**; Registrierung Deutsches Register Klinischer Studien: DRKS00006013

1. Goal of research project

The aim of this controlled, randomized therapy study is in patients suffering from bipolar Disturbance Suffer, currently largely symptom free and stable medic · tare so that they are in regular psychiatric treatment **(1)** the effectiveness of an adjuvant specific psychotherapy (Specific Emotional-Cognitive Psychotherapy = **SEKT)** for relapse prevention a supportive, informative psychotherapy aimed at self-control and self-help (emotionally demanding, supportive, psychoeducational psychotherapy = **FEST)** . This contributes to the evidence -based improvement of clinical care for this patient group. The treatment manuals are made available to the specialist public and thus optimize the clinical care of this patient group . Furthermore, health economics and basic science findings to the short- • and Long-term effects of successful complementary psychological therapy are examined. **(2)** Secondary is about Mechanisms of action (mediators) on neurobiological, emotional and cognitive level one successful Psychotherapy (e.g. SEKT) at Bipolar Disturbances to recognize, to differentiate and to limit. With it become more optimal Treatment decisions and more specific Interventions. Aadditional Research objective is, **(3)** Characteristics to identify, the already _ var Start of treatment later successful van fewer he-
, successful course of the disease allow to distinguish. The testable ones On the one hand , predictors or moderators are neural abnormalities (structural) or the initial effects in the mental control , reward and emotion network (amygdala, ventral striatum, mediale prefrontal cortex, temporo-parietal cortex, subcortical structures) in social-cognitive and emotional tasks (functional). Furthermore, the number of episodes, age of onset, severity of the (last) episode, sociodemographic factors and social functioning level are considered predictors of treatment success (Probability of a relapse, time until relapse) examined.

2. Recruitment plan

As part of this project ("A2Bipolife") should 300 in total patients (intent to treat sample) between 18 and 55 years old the diagnosis of bipolar disorder at 9 clinics - centers (university clinics for Psychiatry and Psychotherapy) can be examined and treated. We expect to "screen" about 500 patients, ca. 360 patients then meet the inclusion criteria, but ultimately continue 15% drop out of the most diverse reasons after all not on the will take part in the study.

The study centers (here with code) have declared (declaration of intent) the following number of study participants to recruit:

01	Tuebingen	50 patients
02	Dresden	30 patients
03	Hamburg	30 patients
04	Berlin	40 patients
05	Frankfurt	40 patients
06	Marburg	40 patients
07	Bochum	40 patients
08	Goettingen	20 patients
09	Munich	20 patients

3. Study Requirements

3.1 Inclusion criteria

In study participate can

- Men and Women
- in the Old between 18 and 55 years
- Diagnosis of one Bipolar Disorder (Bipolar I or Bipolar II) after DSM (296.XX or 296. 89)
- currently in psychiatric treatment/care
- individually appropriate medication after S3 guidelines (mood stabilizer)
- stable (since at least 4 weeks) remitted
- affective episode through clinical relevant cut off score (QIDS – C< 0, YMRS<12) are defined (the Self-assessment measures are not as limit used, however is at strengthen Discrepancies between Self and Third-party rating CONSULT WITH the local study management necessary)
- at least one affective episode during he last 2 years
- and give informed consent (to the Investigations as well as for treatment - development phase including random assignment) for study participation.

3.2 Exclusion criteria

In the study not participate can

- the Age limit (<18, >55 J.) Above - or. fall below
- one missing standby to intake the van medical Page as, individually appropriate assessed medication (usually long-term medication) show
- one existing current or. to the rear (< 6 Months) - outpatient Receive psychotherapy guidelines
- acute suicidal are (Assessment criteria- please refer Section 8)
- a schizoaffective one (295.70) schizophrenic (295.XO) or others psychotic disorder (295.40, 297.1, 298.8, 297.3, 293.XX, 291.XX, 292.XX, 298.9) Suffering psychotic characteristics within more affective episodes place no exclusion represent)
- one antisocial personality disorder (301.7)

- priority at a borderline personality disorder (301.83)*
- a substance addiction (303.90, 304.X) with exception of nicotine (305.10)
- the last 6 months (longer to the back , Substance addictions that are currently and have been stable for 6 months are permitted)
- a Substance abuse (305.XO) exhibit, the the Participation in the study endanger could*
- an IQ < 85 (MWT-B)
- do not give consent or withdraw consent to participate in the examination and study

* Consultation with local Study management necessary

3.3 illness events, setbacks, deteriorations

The occurrence of a new affective episode (depressive, hypomanic, manic, mixed), inpatient psychiatric treatment or the necessary adjustment of the psychopharmacological psychiatric therapy during the study period (18 months) or discontinuation of long-term medication do not provide any Exclusion criteria represent, but are counted as a study event (see below). Patients therefore remain in the study, no - men on the study interventions and the examination dates part.

3.4 Psychiatric primary care (DEW)

Study patients are located as entry requirements in medical /psychiatric treatment (TAU - treatment as usual) and take medication, deterioration, possible hospital stays and necessary changes in the treatment plan cared for. Should patients not have a psychiatrist and receive medical care, then this will be provided by the psychiatric (study) university clinics and polyclinics at the locations mentioned (at least for the study period from 18 months).

4. Therapeutic Interventions (study therapies)

In addition to TAU (medication, psychiatric care), suitable study participants are randomly selected one of the two study psychotherapies **(SEKT, FEST).**

4.1 Specific, Emotional-cognitive psychotherapy (SEKT)

At the **SPECIFIC PSYCHOTHERAPY (SEKT)** acts it itself around one new, in this still in shape not sufficient examined form of therapy, the targeted on patients with one Bipolar disorder is tailored. Preserved cognitive-behavioral elements are preserved (cf. Meyer & Hautzinger 2013) supplemented by metacognitive and emotion regulating Interventions to relapse prevention offered and with the those affected trained. We go of that out of, that particular

Skills are needed, around the instability in the Mood and drive, everyday stresses and signs begin to influence episodes. Ober in total 4 all day Therapy workshops (modules), distributed Ober 4-5 Months edited and Obt one group from those affected (between 4 and maximum 7 Participants) learn these skills under supervision (1-2 therapists); which have proven to be helpful in clinical experience. This includes (1) education and information, (2) mood and activity observation, (3) everyday life and sleep-wake rhythm, (4) social and interpersonal skills, (5) cognitive and metacognitive aids, (6) emotion regulation techniques, • p) Stress regulation and Self-care as well as (8th) Handling with early warning signs, crises and emergencies. Between the full-day therapy workshops, participants are given tasks and Exercises. Linter other become she in addition stopped, Observe your mood every day and keep it in a mood diary (minute sheet) in writing to hold on. For this become she possibly. through the therapists electronically (email, SMS).

SEKT:

4 therapy days (from around 9 a.m. to 5 p.m., approximately every month), as an open group with exercises and numerous materials . Breakdown into one Above all Sessions constant general module as well as individual focal points per session (modules 1 to 4).

General Module: mindfulness, Introspection and tasks edit

To beginning each therapy day will the first both Hours on it used, to carry out mindfulness exercises with the participants Discuss exercises and tasks since the last therapy session , evaluate self-observations and, if necessary, repeat and deepen topics from previous modules.

module 1: psychoeducation, daily structure, Life balance

Psychoeducation with Exchange of personal experiences, communication of basic knowledge about bipolar disorder, explanatory model and knowledge about medication, development of the personal disease progression (“Lifechart”), Connections between events, behavior, symptoms recognize, everyday life, daily structure, sleep-wake rhythm, Alcohol and drug consumption.

module 2: Relatives normality, early warning signs, behavioral changes

What is "normal/abnormal" in • · well-being, thinking and behavior; Identification of early warning signs and investigation countermeasures, development of social and communicative skills through role play and exercises, problem-free interpersonal, professional, financial or other issues associated with the disease problems.

module 3: cognition and metacognition

Connection of thinking and emotions/feeling/behavior, Recognizing typical formal thought disorders (pondering and racing thoughts), dealing with them Distorted thinking (mental filter, exaggerated generalization, attribution style, exercises, reality testing, alternative thought, metacognitive Interventions.

module 4: emotion perception, Emotion regulation
Non-judgmental Perceiving and naming feelings, acceptance and tolerance, effective self-support in stressful situations, emotion regulation through cognitive reassessment and distraction.

Meta-analyses show that psychotherapy with Most of the elements mentioned are helpful, prevent relapse and alleviate episodes of illness. The psychotherapeutic principles mentioned in the S3 guidelines are implemented :

- psychoeducation, Preparation on Crises and Emergencies
- Introspection by mood changes, events, behavior, thinking
- reflection of expectations and scales be
- Financial support from competencies for the self-management from mood swings and Early warning signs
- normalization and stabilization from sleep-wake and social Life rhythm
- Stress management and activity management
- increase the Self-efficacy beliefs

New are the format of therapy and training days at monthly intervals, as well as the inclusion of emotion regulation and metacognitive Strategies.

4.2 Emotionally demanding, Supportive, Psychoeducational psychotherapy (FEST)

The informative one, less so strong structured and not targeted on Relearning aligned , the Experience activating, **SUPPORTERS PSYCHOTHERAPY (FEST)** can as general , less specific Treatment designated become. FEST (oriented at dem BSP/SYSP Manual - Hautzinger, 2009) primarily supports the Personality, the with each • - available resources and the Self-control (Self-control potentials) . It includes the general impact factors of psychological interventions, which are assumed to contribute largely to the success of psychotherapy. This includes social and emotional support, reinforcement, clarification of. Motivation and goals, activation of emotions and your own resources, information and knowledge exchange. In a total of 4 full-day workshops (modules), spread over 4-5 months, a group of those affected (4-7 participants) work on their experiences with the illness under the guidance of 1-2 therapists and help each other, To achieve or strengthen self-control over symptoms, instability, uncertainty and progression. An important goal is for patients to better understand and cope with bipolar disorder and the medication can handle her. The exchange of personal experiences with the disease and the medication - on as well as the personal ones management and processing strategies in the Focus. The role of therapist is less structuring, as rather, helping, allowing, occasionally advising, but primarily the interaction between challenging the participants. There it Hints give t (Meyer & Hautzinger 2012), that supportive, Although general psychotherapy is helpful for bipolar patients, it is considered a proven treatment in this study and active alternative to (new, specific, above, complex) sparkling wine set on. The only therapy material provided here is a text (guide) on psychoeducation (Meyer & Hautzinger 2013) approved and at all Study patients handed out.

Favorable therapist behavior:
Attention to and primacy of affects (emotional experience), positive appreciation and Empathy, Verbalization and Clarification, Specify from emotions, calming, normalizing and supportive insurance, patience, time, allowing awareness, endurance, determining patients the contents and the course of the days, affection, Eye contact, recognition and praise, address strength and ressources, transparency. Clarify and specify, ask interested questions with reference to affect. Summarize with focus on emotional experiences, optimistic, never giving up, hopeful.
If participant interact, do not intervene and let them interact.

FEST:
4 Therapy days (respectively from around 9 a.m. to 5 p.m. , approximately every month distance), as an open group, in terms of content are all days even designed, it exists only one optional module Psychoeducation, the at Requirement be used can.

Optional module Psychoeducation: Use of this present materials, reference to "Advice" text, Elements of this module can be discussed on each of the four group therapy days. It puts not an independent one Module for a whole therapy day, but is carried out hour by hour during the therapy days, if requested by the Participant Need is expressed. Contents can be: Personal experiences with illness, life chart (course of illness), psychoeducation, imparting basic knowledge about bipolar disorder , explanation (Causes), knowledge about medication, compliance, read guides. The personal questions and concerns of the participants are also addressed received.

Therapy modules 1, 2, 3, 4: Exchange Ober personal Experiences with Illness, Experiences and adventures over the last few days and weeks (last month), use concrete examples from the participants, encourage exchanges on topics among each other, topic suggestions could be: day-to-day planning, sleep-wake rhythm, impatience, impulsivity, impact of the illness on the family /relationship, work activity, other illnesses .

Diagrams from and use of the information brochure (guide) are permitted, but no further materials, no exercises, no self-observations, no tasks.

4.3 Personal Interview

So that patients can join each module (therapy day), first one One-off **preparatory, individual session** required. This preparatory meeting earth van _ den ever - is carried out by a temporary group therapist and is intended to introduce the new study participants to the therapy offer, based on their personal medical history and the personal experiences Attachment points to SEKT or. FEST be worked out. It should furthermore Overview of the procedure and the contents are

given. In the preparatory Meeting takes place at SEKT also provides instruction in the Self-observation and the concept of mindfulness as well as the crisis and Emergency contacts. Should it e.g through long Access difficult be, a personal interview Organizing a place is a preparatory task Telephone calls with the same content are also possible. Self-observation forms (sufficient until the first day of therapy) should then be sent to the study participant by post.

4.4 Handling missing times in group therapy

It can happen, that pat. miss appointments. At two parallel the groups at the Study Center or temporally favorable Conditions (e.g patient If he misses the first module he was supposed to take, appointments can be made up depending on the possibility. The Therapy may only occur between t 1 and t2, so in the time of maximum 6 months from t1 take place. It is also possible that patients Group of a fur she accessible others study center visit. Also supplementary course days in which various modules are repeated in a compressed manner for patients who were missing corresponding modules or individual individual crash courses within the framework of which within 2-3h the essential contents of the missed module are to be reworked on the patient, · are possible.
In the FEST condition are the patients due to the session-spanning the contents of the therapy are clear more flexible, so that missed Meetings easy in connection to the regular last meeting rescheduled can be.

5. Measures of success

There the . Goal the adjuvants, specific psychotherapy is, the relapse rate, ie the To prevent or postpone the recurrence of an affective episode in the context of bipolar disorder, the main outcome measure is *the* occurrence of a new affective episode (depressive, manic, hypomanic) defined according to the diagnostic criteria (DSM 5) or by hospitalization determined based on the affective episode .

Through the accompaniment of everyone Study patients Over a total of 18 months with several measurement pointsn further measures of success *(secondary outcome)* can be recorded or calculated: days until to one new episode or. between episodes, days in the hospital due to affective episodes, days of absence and incapacity for work, health costs incurred, changes in medication, medication compliance, undesirable effects of medication or psychotherapy, social and cognitive functioning level.

In the study patients who take part in the magnetic resonance examinations: BOLD values (fMRI) in particular the amygdala, ventral striatum, temporoparietal cortex and medial prefrontal cortex during emotional and social-cognitive tasks.

6. Course of study

6.1 Patient registration

Possible Study participants report yourself, are referred by their doctor, a clinic, psychotherapist, through other patients, through self-help groups, through the German Society of Bipolar Disorders, through the Internet presence (homepage, social networks), through press releases or through information events and the information brochure. Registration is usually done by telephone or email. It is neither in the interest of those affected nor of the study to select patients who are obviously "unsuitable". Study center too order, several hours of initial examinations carry out in order to then determine that very obvious Exclusion criteria present. It should therefore already be on Telephone fans (those affected by email should be called and spoken to in person!) should be made aware of the essential framework conditions of the study and the information on the website. The following registration information should be collected over the phone (see **registration form):**

- Registration date
- Surname, Address, Phone, Email
- Old (Birth date)
- bipolar sick since When, previous Episode count
- current Symptoms or. disease state
- current psychiatric/psychotherapeutic treatment
- medication
- first Examination date

If a person is already clearly violating the inclusion criteria (not bipolar, current psychotherapy), then this should be communicated openly, but the patient should still have the option of participating Group therapies (see section 6.8 Procedure for unsuitable patients) or Refer to other treatment options (e.g. at appropriate outpatient clinics or in private practice). It is further point out that with the invitation to the initial examinations no Decision about the Study participation connected is . First the Results of the initial examinations combined with the express consent from the patients decide whether the study is suitable and accepted. 1st with disproportionately long waiting times (> 2 months) for the start of study therapies to right - nen, then that should also be direct and already mentioned when registering become.
In the Connection to the The patient receives a cover letter after the telephone call Appointment confirmation the dates for the initial examination, information and declaration of consent - physical examination and study participation and the SKID-11 screening questionnaire sent , with the request the Read the documents carefully and the Bring the komplette questionnaire to your first appointment.

6.2 Initial examination (t1)

At the beginning of the study, detailed psychodiagnostic tests will be carried out with each

participant **A gang investigation** (part = Baseline). These investigations include various interviews, various tests (sometimes on the computer) and questionnaires. The initial examinations consist of a maximum of **2 appointments** , each lasting 2-3 hours. These should ideally be within a distance of be carried out for a maximum of one week. Rest breaks during an examination appointment are of course permitted. The time interval between patient registration and initial examination is unproblematic. There should be a maximum of 4 weeks between the initial examination and the start of therapy . Is it extended? period of time by more than 4 weeks, should be euthymic Condition anew by means of the current Symptom measures (QIDS-C/SR, YMRS and ASRM) should be checked and if symptoms worsen, a state of stable remission should be awaited. The reoccurrence from Illness episodes using of LIFE Interviews does not need to be checked at this point, as any illness episodes that may have occurred in the meantime at t2 be recorded.

Before the initial examination is carried out, the patients are informed about the necessity, benefit and purpose of the initial examination (see **information on the initial examination)** and their consent is obtained in writing (see **consent for the initial examination).** One version becomes dem given to patients, one more Version Upper Study director placed in the patient file.

The following **Instruments** come while the initial examination:

- Basic documentation (sociodemographic Information, disease-related information , psychotherapeutic history and current situation, patient code)
- Structured clinical interview for the diagnosis of psychological disorders according to DSM (SKID I and 11)
- Semi-structured interview for the clinical assessment of the extent and severity of depression Symptoms based of “Quick inventory More depressive Symptoms (QIDS-C)"
- Semi-structured interview for clinical assessment of the extent and the severity of manic symptoms based the “Young mania Rating scale (YMRS)"
- Semi-structured interview to assess the course of depressive and manic symptoms as well as other psychopathological abnormalities using the " Longitudinal Follow-Up Evaluation (LIFE)". Relation on the last 6 Months
- External assessment (clinical Assessment) the level of social functioning based on the last month the "Global Assessment of Functioning (GAF)"
- External assessment (clinical Assessment) different functional and areas of life of “Functioning Assessment Short Testing (FAST)"
- Self-assessment more current more depressive Symptoms based of “Quick inventory Depressiv symptoms (QIDS-SR)"
- Self-assessment more current more manic or. more hypomanic Symptoms based the“Altman mania Self-assessment scale (ASRM)"
- Measurement of premorbid general intelligence levels of "Multiple Choice _ l Vocabulary Intelligence Test {MWT-B)"
- Worksheet entry and Exclusion criteria to decision making and Enlightenment

BASELINE (tI)	INTERVENTIONS	FINAL MEASUREMENT (t2)	FO LOW UP (t3)	FOLLOW-UP (t4)
SKID, LIFE	**SEKT** (N=150) or **FEST** (N=150)	LIFE	LIFE	LIFE
YMRS; QIDS-C		YMRS, QIDS	YMRS, QIDS	YMRS, QIDS
ASRM, QIDS-SR		ASRM, QIDS-SR	ASRM, QIDS-SR	ASRM, QIDS-SR
GAF, FAST		GAF, FAST	GAF, NEARLY	GAF, NEARLY
MWT- B		FB THERAPY SATISFACTION	FB THERAPY SATISFACTION	
fMRI*		fMRI*		

Illustration 1: Flowchart.

The mark * points on the *possibility* to participation at others, in elderly with dem A2 project related but separate investigations there. - Study patients can within the framework the Enlightenment talks on this Possibility pointed out become and must at interest at the respectively. Participation in the therapy study is and remains from the moment of participation unaffected by the fMRI examination.

6.3 Clarification and consent to Study participation

At the initial examination **ends personal conversation** with every patient dem . Ober present the A- or. exclusion criteria, with it the Possibility the $tu service participation as well as the essence, meaning and The scope of the study as well as the expected effects and possible undesirable side effects are clarified (see **information for patients).** After this Clarification and answers to all of the patient's questions are signed by the patient and the local study director the declaration of consent (see **consent to participate in the study).** Furthermore, the patient is asked to waive a duty of confidentiality. - dung (see release **from medical confidentiality)** in order to be able to do so Changes of the mood, the medication or in the case one Failure of the patient to appear during the course of the study with his treating doctor . One version is given to the patient, another version is given to the respective person treating doctor to send. A copy of the form becomes Above the Study director placed in the patient file.

The Clinician responsible for initial examination are separately trained and are certified ,

check based on initial examinations the fulfillment the inclusion criteria and the non-existence the Exclusion criteria. She lead also the Enlightenment talk based on patient information. It is also a good idea to use At the end of the initial examination, to discuss with the patient roughly the patient's schedule for the next few months (planned vacations, hospital stays), to the Start of the Group therapy to be able to plan better afterwards.
The decision of inclusion a person into the study follows Inclusion and Exclusion criteria, the signed consent of patients. Only then takes place request for Randomization to the central study management by the local study management.

6.4 Patient Pseudonymization

At the moment the pseudonyms are used for the study participants by an independent Data trustee (identity management system) generated and there along with the identifying details of the study participants are stored. Alie research data Study participants Become just together with this one pseudonym in the Study database (secutrial) saved. The input the Research data he follows also Ober the pseudonym. The link to the personal data is only possible with the data trustee. The use of pseudonyms ensures that personal data and research data are stored separately.
Patient-related communication should also be included the Institute for Medical Informatics Gottingen (MI) exclusively This pseudonym can be used. Sub-projects of joint project Bipolife away everyone Study participants only one ID owns. This must only take place after the patient has given their consent gave his consent to the initial examination.
The Organisation for; society for; party for scientific Data processing mbH Gottingen (GWDG) functions as a data trustee. This is for the assignment between the identifying oats and the pseudonyms used in the databases. Between the clinic representative and the spokesman of the joint project Bipolife will enter into a cooperation agreement for this purpose closed.

6.5 Randomization

After signing the declaration of consent to participate in the study and the consent of the local study management on the correctness of the study inclusion become patients by Coincidence on one the both Study therapies (SEKT or FEST). The Purpose
this approach lies on the one hand, that influence of examiner (e.g. bias) on the assignment of a treatment and thereby on the study results. On the other hand, this ensures that known and unknown influencing factors are present in both treatment groups are evenly distributed.

Random allocation is carried out by the study center, which uses the randomization lists created by the independent biometric center (Numerus Ltd). managed. Suitable and consenting patients will be reported by the local study management by fax/electronically (telephone). The randomization form can be completed by the study diagnostician but must be signed by the local

study director or a representative to ensure study inclusion agreed with the local study management. The local study director will find out from the study center (by fax/electronic and letter), which treatment Study patient should receive and which one lnter vention condition he/she was assigned. The randomization lists are only accessible at the study headquarters and only by the study director plus a representative.

To randomize, a study patients will be following **information** the Studies reported/requested to the central office or reported back as a protocol:

cipher (employee code)	**01** (for Tubingen) **101** (for Study director) or **303** (for study diagnostician) 1 011101
Cipher (patient through identity management	_____ system generated code I N. N

Old in years	e.g.	27
Gender	e.g.	m
Number Episodes	e.g.	2
last Episode	e.g.	04/2014 (Month year)
Date	e.g.	04 . 10.2015

Signature (local study director or representative Study center)

_____

Randomization result SEKT or FIRMLY
Date
Surname, Signature {central Director of Studies out of study center)

The by Coincidence selected intervention becomes directly (electronic, by phone) from the study center at the local Study management reported. The local Study management informs the study therapists (never the diagnosticians) of the decision and creates a patient file at and organized the more Proceed. The patients werri either also
by the local study management or by the study therapist as part of the appointment for the preliminary discussion about the selected intervention.

6.6 Follow-up examinations (t2, t3, t4)

Depending on the timing of the interventions, Time of initial examination { tl) viewed after 5 to max. 6 months (t2 = end of treatment, but independent of that, if the therapies have been completed), after 12 months {t3) and after 18 months {t4) a new examination (interview, questionnaire) of the participating patients. This end and Progress measurements are used for assessment the effectiveness and Safety of the treatments as well as to analyze the mediators and moderators of the forms of treatment used. The employees who carry out these investigations (to tl, t2, t3 or t4) are not informed ("blinded") about which form of psychotherapy individual patients are receiving or have received. The patients are expressly instructed, not about the interventions received to speak. The investigating ones Diagnosticians also inform patients of this at the beginning of the t2, t3 and t4 measurements. These follow-up examinations are in the Usually lasts about 2-3 hours to carry out the appointment. Rest breaks are provided.

As at the initial examination the following interviews and instruments the final measurement (t2) and the two catamnesis appointments (t3, t4) for application:

- Semi-structured Interview to assess the course of depressive and manic symptoms as well as other psychopathological abnormalities using the " Longitudinal Follow-Up Evaluatio _ n(LIFE)" with reference on the last 6 Months
- Semi-structured fnterview for the clinical assessment of the extent and the severity of depression Symptoms based of "Quick inventory More depressive Symptoms (QIDS-C)"
- Semi-structured interview for clinical assessment of Extent and the Severe manic symptoms based on the "Young Mania Rating Scale (Y.MRS) "
- External assessment (clinical Assessment) of social Functional level based on last month using the "Global Assessment of Functioning (GAF)"
- External assessment (clinical assessment) of various functional and areas of life of "Functioning Assessment Short Testing (NEARLY)"
- Self-assessment current depressive symptoms based on the "Quick inventory of depressive symptoms (QIDS-SR)"
- Self-assessment more current more manic or. more hypomanic Symptoms based the "Altman mania Self-assessment scale (ASRM)"
- questionnaire to Therapy satisfaction (exclusively to t2 and t3)

How be previously this surveys or Assessments are largely direct at the Computer recorded and sent to the central data center (Ml.) after completion of the study (and in this case with the consent of the study participants). Gottingen) Chief Ambassador.

6. 7 Schema for Participant

Everyone interested patient passes through first the registration, the initial examination and the enlightenment discussion (see relevant sections).
Decides yourself patient against Participation at the study or become the Inclusion criteria eff not Fulfills or. lay Exclusion criteria var, then becomes a more interested Patient not to the study patient. The Patient record will be closed and the Patient data will be deleted (it be because, a Patient expressly allows this Further use the as part of the Initial examination collected Data - please refer in addition patient information).
A patient becomes for the study patients, if he after Initial examination all Inclusion criteria met, no Exclusion criteria present and his expressive approval to
· Study participation with all associated surveys and interventions erte . ilt . With who then occurring Randomization is a patient in the study recorded, completely independent of whether he appears at further appointments or in the interventions and surveys cooperated.

Temporal Flowchart:	by telephone, by letter, electronically
Registration	First Contact (telephone)
	Initial examinations (2 Appointments of 2-3 hours)
	Randomization
	Interview
	Greetings, brief information about the study, answer already existing questions, contact details, Clarification the fundamental Fit interviews, assessments, Questionnaire, Information, consent, preliminary appointment for final examination through Study Center Forwarding to local study management ll"!formation van patient and Study therapist individually by study therapist FEST or SEKT group therapy

EFEST or SEKT Group therapy

Month 1

Therapy day 2

Therapy day 3

Therapy day 4

Month 5-6 Month 12 Month 18

Final examination	interviews, assessments, Questionnaire
{meeting to 2-3 H)	scheduling for Catamnesis 1
Catamnesis 1	interviews, assessments, Questionnaire
(meeting to 2-3 H)	scheduling appointments for catamnesis 2
Catamnesis 2	interviews, assessments, Questionnaire
(meeting to 2-3 h)	thanks, adoption

Typically a study patient is in the study for 18 months. During the first 6 months, initial examinations and interventions {over 4-5 months) and final examinations instead of. Follow-up examinations (catamnesis) are for Month 12 and month 18 - planned.

6.8 Proceed at unsuitable patients

If the patients who were excluded during registration or the initial examination are still interested in the Participate in therapies, as well Therapy places are available in the current groups, they can be treated as therapy participants outside of the study protocol at the Participate in therapy groups. It will recommended that these patients at least Investigations at t1 and t2, but this is subject to the decision of the respective local study center.

7. Serious, unwanted events (SAE)

Adverse Event (AE, German: undesirable event) is every adverse event that occurs to a data subject participating in a study, but which is not necessarily causally related the examinations or treatments stands. In this study are the following AEs conceivable:
worsening of symptoms , Occurrence of new symptoms, occurrence of passive or active suicidal thoughts or Suicide planning, problems in the patient-therapist relationship, Problems in the therapy group and other negative incidents (such as for example a flu A broken leg , a sports or traffic accident , etc.).

Seriously (serious) adverse Events (SAE) are defined as unwanted Events that : the death of one person have as a consequence, directly are life-threatening (e.g. suicide attempt), a hospital stay or the Renewal one hospitalization result in a permanent or serious Disability or invalidity for fol - have had .

A SAE is in each case to document. This applies away dem first Contact for the Studies center , so also already for en Period between registration and the Initial examination. The SAE protocol will Part of a study patient's medical record. A copy of the SAE protocol is sent to the study center.

7.1 determination of the causal relationship

The Decision the **causality ,** so if a SAE the Consequence the Study investigations or the study treatments is **determined** by the study staff involved (diagnosticians, therapists). sure . An assessment and control, if necessary a correction carried out by the local study management, the study center and if necessary the DMSC.

Any SAE will through the examiner with reference to his causal relationship between dem Event and the study/den Study measures rated.

secure	A unwanted Event , at which one one plausible Possibility exists that it is through the Study participation caused became , which a tight temporal Connection to Study therapy has and in which one Alternative cause unlikely appears .
Probably	A unwanted Event , at which one one plausible Possibility exists that it probably through the Study participation brought forth - called forth became . That - unwanted Event is related to time with the study therapy and follows one known Reaction pattern , but one potential Alternative cause can be knows be.
Possible	A unwanted Event , at which one one plausible Possibility consists , that it probably through the Study participation brought forth - called forth became . The unwanted Event is in time

	context . with the study therapy , follows but no one known Response patterns . An alternative cause is exists and appears more likely or it lies a significant Uncertainty regarding the cause of event before .
Unlikely	An unwanted one event , at which one . only a distant connection between dem event and the Study therapy exists . Other circumstances How e.g simultaneous diseases , one progression or a acute attack the underlying disease / one Pre-existing illness or one Response to concomitant medication seem that reported Event plausible to explain .
No context	A unwanted Event , which . it not in one plausible temporal context to Study therapy stands and for which one plausible Alternative cause consists .
Not assessable	Inadequate Oaten, the not to Evaluation sufficient .

The causal correlation between dem Event and the study/den Study measures can be assumed to be positive if they are assessed as follows:

,,secure"

- ,,probably"
- ,,possible"

Should there be a connection between the event and the study /the study measures in accordance with the above definition, it is a "serious adverse study-related event".

7.2 Definition of severity

The rater use the following definitions of the severity, to the maximum

Light:	Unwanted things Event, usually temporary and in the Generally not affecting normal activities of daily life .
Mediu	Unwanted things Event, which normal Activities of daily disrupting life impaired . Unwanted things Event, which to inability to work leads and normal activities of daily life
Difficult	

7.3 Procedure

In the Cases one SAE are the following **steps and Procedures** to follow:

- The **Serious -Adverse Event Form must** be completed by the local study director and submitted within from 24 hours after becoming known SAE at the Study Center be faxed. If the study patient was hospitalized , if possible , the date of discharge - Note the solution and record whether the patient continues to participate in the study. Hospitalization - tion alone is not Reason for the exclusion of a study patient .
- The **report is checked by the study center** for completeness and formality Plausibility checked . At become a need Queries placed and tracked. It must Care must be taken to ensure that personal data is only used in a pseudonymized form - dung the identification code of the data subject is transmitted and, if necessary. from there given become.
- It he follows the **Evaluation,** if the unwanted Event is serious , a Causal too - related to the study therapy or. -examination is at least possible , that the event is unexpected, Measures to protect the persons affected from immediate - to avoid danger .
- Formally still not completed **Events are tracked.** If the SAE still ongoing (e.g. the patient remains in the hospital), the sends Director of studies a white teren Report to the Study Center a week after Initial report. Finally, a final report is sent to the study center, when the undesirable event has been resolved.
- The **DMSC** will from the Study Center informed and around Assessment requested.
- Has itself based on reviews highlighted, that a Suspected case SAE then the study center will inform all local study directors immediately , but at the latest within 7 days of becoming known **.**
- The respective Study centers should furthermore the Regulate of their local Ethics Committee - nen consequences and them the required Information i.ltransmit .

7.4 Suicidal Tendencies

Suicidal tendencies is a common Theme at patients with one Bipolar Disturbance. Therefore needs this Theme the constant Attention from everyone study staff, the in the Contact to the Study patients stand. Indicators for Ir acute Suicidal tendencies are:

- patient distant itself not from suicidal ideation, Suicide attempt
- Patient experienced urgent Suicidal thoughts
- patient works extremely hopeless
- patient has no Future perspective
- patient is social isolated, has itself in last Time increasingly withdrawn.

- patient has Conflict, the to Suicidal ideation/attempt led , unsolved .
- patient Reacts pronounced irritable/aggressive/agitated, no more sustainable rapport
- Pati nt has heavy depressive upsets, Possibly. with depressive Delusions
- patient has a or several Suicide attempts in the prehistory
- patient has a Suicide arrangement met, the difficult to find, impossible might
- patient undertook Suicide attempt with harder method or has appropriate ideas
- patient shows lack Impulse control

Examples for indirect Comments from Intentions to commit suicide:

"I'm sorry cases each only still to the burden. - Mine Position becomes itself never improve. - I liked that that everything stops. - My whole life is pointless. - There are also another way.

Perhaps see we us not more. - Life She probably" instead of ,,On "See you again."

Determination the Risk level

High Risk	If the patient the active Wish has, Suicide too commit, the Self control or the external Support (e.g family, friend) for Safety precautions missing, then is the introduction to one Psychiatric clinic necessary and must from the study staff caused become
Increased Risk	If the patient suicidal Plans has, but without the active Wish , sui • cid to commit this one urgent situation represent, the to could become an emergency . The patient should be seen within 48 hours - man see and should knowledge, where · Emergency assistance to receive is.
Low risk	If the patient does not have suicidal plans and no active desire suicide to commit, then became this situation as low Risk assessed. Further measurements at this time are not necessary. It should with every contact one Change this status geach tet become

The risk of suicide, the course of the conversation and the procedure must be documented in writing. The three-stage risk scale and the risk factors mentioned can be used to clarify the severity and to document it be used.

Conversation and Handling with suicidal People:
Targeted Talk about suicidal Danger leads in the rule discharge and to the open - solution to suicidal isolation and confinement. The patient usually learns that his distress signals are being accepted and that, in principle, there are options for help given are. There is no fear of aggravating or causing suicide by addressing suicidal tendencies.

Generally should the following Points get noticed :

- Time win and possibility of conversation and -create atmosphere . Short-term discussions make possible.
- Openness, Understanding, direct Address, but also patient not for the Conversation
- force .
- Serious take, not protect or trivialize, but also not dramatize.
- acceptance and Understanding of suicidal behavior as Expression more spiritual distress.
- Depathologizing suicidal behavior, objectifying it, reducing shame, sin, avoiding judgment.
- Communicate hope: Suicidal tendencies are (usually) times of crisis in life, during a period of illness, the but also can pass again. Show in hindsight itself often other Troubleshooting .
- Bindings (if available) address, also Values.
- Inclusion more important caregivers (provided positive experienced), contacts produce.
- The · suicide risk should in the Course regularly checked become.
- Complete shorter term Contracts of therapists with the:n patients. Fit She these over and over again.
- patient is Ober those taken for his protection decisions as much as possible Exactly informed and involved in the decision as far as possible.

At everyone Measures are the legal Regulations to observe.

Legal Aspects:
Suicide itself is unpunished in Germany as an expression of a person's right to self-determination . suicidal tendencies a Emergency. Anyone at the scene of an emergency must help. Non-prevention one suicide attempt, if this was foreseeable and could have been avoided is failure to provide assistance and had:

a) Criminal law Consequences: Failure to do so assistance, negligent act/ Bodyver last
b) Civil law Consequences: Claims for damages
c) Professional law Consequences: Danger the Approval

In order to be able to provide help, in the case of acute suicidal tendencies, it is permitted to take the person committing suicide with them use appropriate means to deter him from his plan (call in relatives or significant others, call an emergency doctor or the police, etc.).

Measures at more current Suicidal act:

Each Study Center has to ensure that procedures for a current suicidal act in - implemented. Examples of this are:
Immediately with First aid measures and possibly. with Resuscitation measures begin and at the same time the Emergency doctor request.
1st the Emergency doctor arrived, he takes over the Resuscitation. The criminal police are informed by a doctor.
At Conversation with the Criminal Investigation Department only Ober Facts e.g diagnosis et al Information ge
ben, no speculation Ober the Backgrounds of suicide hire.
The Medical records will too at the request of the Not handed over to police! This requires a written request, if necessary with the involvement of the legal department.

8. Dropping out of studies or. Exclusion from studies (drop out)

It gives some reasons, for what reason a patient the study prematurely completed:

- **Patient decision:** The patient can decide, his consent to increase and not continue the study. The local study director will be in collaboration

with the diagnosticians or. therapists the Reasons for this Decision clear up. The Pati will be removed but also informed that he Reason does not have to be given. It is documented whether the patient insists on deletion of the data collected from him in the study documentation. The information that when and in which arm a patient was randomized and that and when he withdrew his consent must be retained in the documentation.

- **Lost to Follow Up:** If a patient multiple at Therapy sessions missing and the study final diagnosis or. Study management a patients not to contact can, becomes the Pati ent as 'Lost to follow up'.
- **Additional exclusion criteria:** As soon as is certain, that a Exclusion criterion before lies, has to be patient out the Study excluded can be however possibly. the therapy as well the Diagnostic appointments e.g End lead. This is in the individual case decide.
- **Moving away, long absence:** If a patient moves away before the third therapy module or is absent for a long time (months) and therefore can no longer be present on therapy days and examination appointments, he must be excluded from the study .
- **Medical Decision:** A Dropping out of studies can because of one medical Decision if the study is continued an unreasonable and ethically unjustifiable situation for the patient would mean a burden.

Protocol violation or. Non - compliance are no Reasons for a Exclusion from studies. The intent-to-treat principle applies, which means that every included patient remains in the study and in the analyses according to the principle "last observation carry forward"
procedure becomes.

At Dropping out of studies must the Reasons or. Circumstances and the last status (symptoms) documented_ become. Pulls a patient his No study consent back, should the Follow the course if possible and the patient too the follow-up examinations be invited.
Any dropout will be handled by the study center with Date and stating the circumstances and Reasons documented (see **dropout documentation).**

9. Documentation

Within each study center are the the following files for documentation and archiving of the with related to the study Setting up and maintaining documents.

9.1 Study file

The study file (TMF: Trial Master File) contains the ethics approval, correspondence with the local ethics committee, protocols of the initiation and further control visits (monitoring) as well as the list of study employees (short CV with Qualification, Certification, start and end of collaboration, role in the study).

9.2 Patient file

Even if the interviews, the ratings, the external and self-assessments are carried out on the computer and the associated answers are largely entered directly, a patient file must be created and locked at each study center for each patient examined and **thus kept spatially separate** from the collected patient data preserve. The "blinded" Diagnostician have **no** Access to these patient files. The patient files are kept and managed by the local study management (or representative). The study therapists have access to these files to access contact details and to file the therapy documentation.

The Patient record contains for every patients Ideally in following Re . order:

- Progress plan (investigative and therapy appointments) of Pat. through study
- Registration form
- Copy **or.** Aμsdruck the first Page of Basic documentation (until including point 6)
- signed consents of Pat . . to Initial examination and Study participation
- Copy the signed medical Release from confidentiality
- Randomization form
- Therapy documentation (at latest after the Intervention phase through therapists)
- possibly . Forms to documentation from SAE or Dropping out/exclusion from studies

The Answer and Assessments regarding the others to the four Examination time

The instruments collected from the points are stored and checked centrally (via ecuTrail at Ml Gtittingen) and completed during the course of the study. The (personal) information in the patient file is not allowed the patient data collected at the time of the examination can be brought together. Based on the patient code alone, it is the respective local one Study management possible . H, one Assignment from stored Oaten and personal ln
formationeri to be carried out. Oieser case is however always with the Study Center (total performance
•
tung) and dem DSMC board to vote and only at present one "serious,
unwanted event" (Seriously Adverse Events - SAE) permitted.

9.3 History file for the Diagnosis

It is recommended that a progress file be created for each study patient, which must be kept by the respective diagnostician and contains contact details and documents from the individual examination appointments . This progress file must be handed over to the appropriate diagnostician if there is a change in diagnostician. After completion of t4 and thus the completion of the diagnostics, the from this important content above Head of studies in the to file the corresponding patient file (see above). There too History files contain confidential information
' tions can include, is this always locked to keep.

9.4 History file for the therapy

It becomes recommended, for every Therapy group one History file to create, the from dem respective therapists fears i . st and the contact details as well as the Therapy documentation included and as Memories serve can. After completion the intervention are the from it important Contents in the appropriate Patient record (so) to put down. There also the History files for the therapy contain confidential information can, is these always locked to keep.

First Contact (telephone)

Initial examinations (2 appointments lasting 2-3 hours)

allocation to one of both Conditions ever after Requirement

Features

10. Study Center (overall management)

The Study Center is located on Eberhard Karls University, Chair of Clinical Psychology and psychotherapy at the Department of Psychology (Schleichstr. 4, 72076 TObingen). The overall management lies with Martin Hautzinger and his employees (Julia Zwick and Raphael Niebler). The study coordination, all training (diagnostics, therapists), initiations and controls of the clinical centers regarding compliance with the study protocol are carried out by dart and the Patient inclusions. The Study Center takes over central
tasks such as creation and updating If necessary, adjustment of the study protocol, of the organizational and Process plan (SOP, CRF), contact with the ethics committee and Ethics vote, the organization of (regular) telephone conferences and working meetings of the local study leaders, clarification of central questions and concerns, contact with the DMSC, recording and clarification van SAE , publication agreement, Regulations on external and Internal communication, public relations work, internet presence, homepage, data control and contact with Ml Gottingen, preparation of statistical analyses, evaluations and publications.

<u>Contacts for Study center:</u>
Martin Hautzinger
07071/29-77301 (Univ.), 0179/5443124 (mobile), hautzinger@uni-tuebingen.de Julia Zwick
07071/29-75326 (Univ.), julia.zwick@uni-tuebingen.de Raphael Niebler
07071/29-77181 . (Univ.), raphael.niebler@uni-tu bingen.de

10.1 Local Study management

At every center one local Study management (see employee **list).** to name. This person or their representative is in close contact with the study center. Find it between the local Study Directors and the Study Center . e r: regularly telephone conferences and Voting contacts (via email) as well occasional direct work visits take place.

The task the Study management belong:

- Guide and Update the Study file (TMF : Trial master file)
- Line and Instructions of local clinical study center
- recruitment and Measures for extraction van Study patients
- observance of study plan (SOP) and the Schedule (CRF)
- organization and control the Registration and Initial examinations
- Decision Ober the Admission to study one patients
- Randomization contact to Study center (randomization protocol)

- implementation the Randomization decision
- Creation and Guide van Patient records for every patients
- Directions, control (supervision) the Study diagnostician
- Directions, control (supervision) the Study therapists
- Measures to Aversion van SAE

10.2 Data and Safety Monitoring Committee {DSMC)

The DSMC is a international board, that the entire Bipolife joint project different Subprojects available stands. For this study is the Function: Bera - tung during the course of your studies, Support with difficulties and unwanted ones events, monitoring compliance with the study protocol, suggestions and assessments for data management , assistance with evaluation questions, utilization of later results. The members of the DSMC are:
Joerg Hasford , MD Professor
Institute for Medical IT the Ludwig Maximilian University Marchioninistr . 15
81377 Munich D

Steve Jones, PhD Professor
SEKTtrum Centre for Mental Health Research
School of Health and Medicine Lancaster University
Lancaster LAl 4YT UK

Thomas D. Meyer, PhD Professor
Health Science Center University of Texas
Deptm. Psychiatry and Behavior Science 1941East Road
Houston, Texas 77054 USA

Martin Keck, Prof. Dr. Dr. Professor und Direktor
Max-Planck-lnstitut fur Psychiatrie
clinic for psychiatry, psychotherapy, psychosomatics, Neurology Kraepelinstr . 2-10
· 80804 Munich

With the DSMC, regular telephone conferences and working meetings take place at one of the study centers instead of.

10.3 Employee coding

All study employees receive employee codes, which are designed as follows: study center code

(see recruitment plan) plus a three-digit code for the employees - speaking of the task category (see below). When changing functions or employees, including new employees, new, ie consecutive numbers are assigned.

Study management	100 (beginning with 101)
Study organization	200 (beginning with 201)
Study diagnostics	300 (beginning with 301)
Study therapy	400 (beginning with 401)

The study management, for example, in eochum therefore receives 07101, the study organization helpers in Gottingen receive 09201, for example and 09202, the study diagnosticians in Marburg receive, for example, 06301, 06302 and 06303, the study therapists in Hamburg receive, for example, 03401 and 03402. Employees leave or new employees are added, regardless of the task category , then new numbers are continually assigned. It is So it is possible , but purely hypothetically, to include 99 study directors at each location, ie for Dresden, for example , 02101 to 02199. Even in the other areas of responsibility, these numbers are never filled out .
The center codes , the Employee codes and later the patient codes will be fi..ir the diverse sen Forms and study materials required for assignment . This is intended to provide additional data protection The invoice carried become.

12. Publication agreement

In principle , publications using the materials and data collected in this project can only be published after explicit consent **Approval of the study center** as well the at the beginning mentioned Those responsible for studies at the study centers take place major fen . Any Texts and Manuscripts are to Release the Study Center or. possibly. one to be set up Publication board punctual (at least 4 weeks before e . i he planned submission) to present. The at the Ml Gottingen electronically Obermitten Study data is available until Diploma the study **not** fi .. ir Publications to Verfi .. confirmation . From everyone this study emerging publication , himself under use from Share -

Appendix
Information to the Initial examinations

The study ,, A2Bipolife - psychotherapy for patients with Bipolar Disorders "
Study management (Sponsor, Responsible person): Prof. Dr. Martin Hautzinger (see above) Dear patient,
You are interested in a scientific one Study of psychotherapy for bipolar patients - to participate . Before you and we can decide whether to take part , we must enter the information - investigations necessary. From our side, there are inclusion and exclusion criteria for participation in this therapy study set. From your side, there is a rightful need for information if necessary for or against participation decide.

Goal the scientific Investigation
In The aim of this study is to determine the effectiveness of a specific psychotherapy (Specific psychotherapy = sparkling wine) for relapse prevention a supportive, self-centered
- steering and Self-help aligned and informative psychotherapy (Clinical Support and Psychoeducation = FIXED). You remain above that entire study time in treatment your doctor. 1hr Doctor is fur the medication, for the treatment of setbacks and crises, also for necessary hospital stays may be required. The examinations and psychotherapies offered are additional and complementary to this (specialist) medical treatment .

Sequence the Initial examination
Var beginning of the study - will be detailed psychodiagnostic Initial examinations with To determine whether our offer is fundamentally valid is suitable for you. This investigation - The tests include various interviews, various tests (sometimes on the computer) and questionnaires . This Initial examinations include most of time **2 Events** van respectively approx. 3 Hours Duration.
- After the initial examination has been completed, a detailed examination will be carried out Talked about where you are upper being, meaning and Scope of the study and the two psychotherapies as well as expected effects and possible undesirable side effects are clarified . As far as possible information we guess You are happy to do so also about the findings of the initial examinations.

After signing the **consent form** You will be invited to take part in the study Using a computer program randomly assigned to one of the two treatments. This procedure - ren will be **randomization** called. The purpose of this approach is, on the one hand, to: Influence of the examiner (e.g. bias) on the assignment of a treatment and thereby on the study results . On the other hand, this ensures that known and unknown influencing factors are evenly distributed in both treatment groups. The probability of one of the two treatment methods to receive is 50% each.

Possible Risks the Initial examination

With the upcoming ones Investigations, surveys, interviews and questionnaires are not risks tied together. There in Framework of the initial investigation on your illness, the illness episodes and others with of bipolar disorder If the associated experiences are discussed in detail , this dispute may lead to temporary stress. Please inform us of any deterioration in your condition with, dam, t we possibly . the investigation can interrupt or end .

Voluntariness the participation

Participation in this initial examination and later Participation in the scientific study is voluntary. You can yours Participation in the initial examination at any time without specifying Reasons to end . This means there are no disadvantages for you!

confidentiality the Oaten

The personal data and information from this scientific Study will be ver treated confidentially . The within the framework of the _ r scientific The data collected in the study are pseudo nymized . that is, without reference or mention of you personal data, such as name, Address, in compliance with the legal regulations of Data protection of study employees after prior instruction and more explicit Declaration of confidentiality, scientific (statistically) evaluated - tet . By agreeing to participate in this scientific study, you agree that you with the recording made as part of this investigation from Oaten to yours health and the findings collected and their evaluation by scientific staff of the study . All persons who have access to your documents or who evaluate the findings and scientific data are subject to data secrecy in accordance with the Federal Data Protection Act bound.
The data management of the entire study (with 9 clinical centers) as well as the entire joint project " Bipolife " (with 7 large studies) are carried out by the institute tor Medical Informatics at the University Medical Center Gottingen . The data is stored in accordance with the data protection concept of the technology and method platform tor the networked medical research1.mg eV (TMF - Tele-Matic platform) only in pseudonymized shape saved. That means , all Study data will be separated of name, date of birth, personal Information (so-called identifying data - IDA T) saved - chert . A more trustworthy data trustee , the nothing with the study and the Evaluations to do has, is without our Possibility of influence for the Pseudonyms responsible.
Data collection and storage is carried out using the professional study software SecuTrial (www .s ecutria l. com) . With SecuTrial Multicenter studies can be carried out become. SecuTrial is GCP compliant (standard : “ Good "Clinical Practice" of the WHO) and is great all international requirements Rules goal Therapy studies and Approvals (FDA Regulation 21 CFR Part 11).

Before use, the database was carefully examined with regard to the questions , interviews and instruments used as well as the functions used tested . insight into Only study managers and specifically designated controllers (monitors) receive the information entered . An export of the data for statistical purposes Evaluation will only to the in the Study protocol described points in time a n d i s reserved for the head of the study or the statistician of the study. The structure, processes and legal Basics of data management become in one separate Data protection concept

detailed explained. She find the already from numerous ethics committees and Data protection concept checked and approved by data protection officers at www.b i po l i fe . de . We would be happy to provide you with a printout of the data protection concept available .

data collected as part of this therapy study will be used for 10 S t o r e d pseudonymously for years (with no apparent reference to you , encrypted) and during this time for scientific - che , never personal Evaluations used. The respective users are scientists joint project, the before through the study director and the security board Chief Inspection and the use of the oats approved became.

In individual cases and according to a random plan, the initial examinations are recorded (tape , video). This is done solely to monitor employees Study management. If this happens, then informs her the respective investigators and get one for that separate ones Consent form a. She are not obligated the recording to agree. The Participation in the therapy study is still possible .

Notice on insurance

FOR No separate insurance will be taken out for the planned examination. However, there is insurance coverage through the University and Clinic Obligatory Liability Insurance.

Prof. Dr. Martin Hautzinger

Appendix
information for the Patients and patients

to study
“A2Bipolife - psychotherapy for patients with Bipolar "disorders"

Study management (Sponsor, Responsible): Prof. Dr. Martin Hautzinger (so)

Very dear patient, very honored Patient,
You have been asked about your willingness to take part in a scientific study to participate in the investigation. Before you make your decision, we would like to give you important information about this study. Please read the following information carefully and contact us if you are unclear or have additional questions.

General information
Bipolar illnesses affect the majority of people Those affected have a chronic course, i.e. H. Episodes of depression or mania occur again and again. The effects of these repeated phases of illness are many times higher than other illnesses such as depression, which also occur in episodes and then subside again. Chronic bipolar disorders that occur repeatedly in phases lead, among other things, to severe psychological, social and family problems and more professional Impairment. In the majority the Cases begins the Bipolar Disorder var 18 . year of life, is lifelong and always requires accompanying, mood-stabilizing medication . While for a few years people were still convinced that the long-term medication with "mood stabilizer" alone medication) In order to be able to influence the course of bipolar disorder and prevent new episodes of the disease, it has now been recognized that medication alone is not enough; Despite long-term medication, setbacks occur and therefore accompanying psychotherapy for bipolar disorders is appropriate for the success of the treatment is increasing. However, the scientific knowledge is not yet sufficient for this assumption. Above all, it is unclear whether The use of specific psychotherapy is particularly helpful and particularly indicated for preventing, weakening or delaying new episodes of illness .

Goal the scientific Investigation
This study is about patients who suffer from bipolar disorder and are suffering from it permanently and stable medicated are, the effectiveness one specific on the training of skills and coping options psychotherapy (= SEKT) to prevent relapses an emotionally demanding, supportive, self-directed and self-help and informative psychotherapy (= FEST).
Included stay above the entire period of study in the Treatment of yours doctor. 1st doctor is for the medication for the treatment of setbacks and crises , also required for possibly necessary hospital stays. The examinations and psychotherapies offered are **additional and supplementary** to this (specialist) medical treatment. Therefore request we van you also the

Permission to contact your treating doctor and about your study participation . hme to inform. At the same time we request She, during the Study phase, that is at least for the next ten 6 months to remain in specialist treatment and to continue the existing medication.

Information to the Psychotherapy procedures

At dem specific Approach **(SPARKLING WINE)** acts it itself around one new, in this shape still not sufficient
. chend examined therapy forms that specifically target behavior, information processing, thought processes and Emotion regulation van patients with one Bipolar Disturbance tailored is. We go of that out of, that certain skills are required, · in order to the Instability in mood and in the drive, the everyday stresses and Signs of beginning episodes influence. In a total of 4 full days (9 -17 p.m.) therapy workshops, spread over 4-5 months, edit and one at the top group van those affected (4-7 Participant) under Instructions (1-2 therapist) particular skills, the itself in the clinic as helpful to R relapse prevention proven have. In addition belong (1) Enlightenment and Information, (2) .mood- and activity observation, (3) everyday life and sleep-wake rhythm, (4) social and interpersonal skills, (5) cognitive and meta cognitive aids, (6) emotion regulation techniques, (7) stress regulation and self-care and (8) coping with early warning sign, Crises and Emergencies. The Expiry of respectively all day Workshops are structured and aims on the acquisition newer, helpful skills for disease management . Between the all day one Get therapy workshops the participants Tasks and Obun
gen.

The supportive, informative, less structured and focused on new learning, the emotional experience and the own Strong-oriented psychotherapy **(FEST)** can be used as a general, Treatment that is less focused on specific skills but still demands emotional experience. FEST primarily supports the personality that exists in everyone Resources (Strongs) and the self-control. It includes the general impact factors of psychological interventions, which are assumed to contribute largely to the success of psychotherapy. This includes social and emotional support, reinforcement, Kia tion of motivation and goals, activation of emotions and own resources as well as information and knowledge exchange. In a total of 4 full-day workshops, spread over 4-5 months, a group of those affected (4-7 participants) work on their experiences under supervision (1-2 therapists). the Illness and help each other Instability, uncertainty and course to achieve or strengthen.
A important goal is that patients better understand and cope with bipolar disorder, including the medication can handle her. In addition, there should be an exchange of personal experiences the disease and the medication take place. The therapist's role is less structuring, rather than helping to express conflicts, problems, doubts and the associated emotions can and together for solutions and help seek.
There in Investigations is occupied, that FIRMLY at Bipolar Patients good Results he brings, becomes it in this Investigation as preserved Treatment and active alternative to (new) SPARKLING WINE used .

Sequence the study

As part of this scientific The study will involve a total of 300 patients between 18 and 30 years old the Diagnosis of bipolar disorder at 9 clinical centers in Germany examined.

To The study will begin detailed psychodiagnostic **Initial examinations** with c o n d u c t e d to you to determine whether our offer is fundamentally suitable for you. These examinations include various interviews, various tests (sometimes on the computer) and questionnaires. These initial examinations include **2 Appointments** lasting 2-3 hours each. At the end of the second examination appointment, there will be a discussion in which you will be informed about the nature, significance and scope of the study as well as the expected effects and possible undesirable effects. As far as possible, we would be happy to inform you about the findings of the initial examinations. After signing You will be given the **declaration of consent to participate in the study** With the help of a computer program by chance of the two treatments (SPARKING or FEST) assigned. This procedure will **Randomization** called. The purpose of this procedure lies, on the one hand, in the influence of examiner (e.g bias) on the Assignment of a treatment and thereby on the Study results to be excluded. On the other hand, this ensures that known and unknown influencing factors are evenly distributed in both treatment groups . The probability, one of the two Receiving treatment procedures is 50% each.

There in this study two different Forms of psychotherapy must be compared **before** starting the study, i.e. before the random assignment to one of the two treatment conditions and before starting psychotherapy, be sure that you are under medical treatment, a mood stabilizing treatment Take medication reliably and regularly and is currently not in an acute phase of illness (depressive or manic episode).

The psychotherapeutic outpatient study options **(SPARKLING WINE or. FIRMLY)** extend Upper approx. **4** months at **4** training days from each approx. 8 hours Duration, in the rule in Distance from Typically **4 weeks** non-working Saturdays.

end and Follow-up examinations

After the end of treatment (around 5-6 months), riach 12 months . th and after 18 You will be examined by staff for months to assess the effectiveness and safety of the treatments (interview, questionnaire). These follow-up appointments each last 2-3 hours and are included According to you agreed upon according to your possibilities and timing.

Those examining you employees are is not informed about it, wel5=which form of psychotherapy you receive or. receive have. they should with these employees too **not** above that Study treatment speak. Also if this Follow-up appointments first in some Months _ n take place and the accurate planning for that still pending, request we She already now around yours Permission to this follow-up examination.

Video recordings the Therapy days

From the therapy days, both at SEKT and at FEST, but also from some interviews (the introductory and final examinations) become tape - or. Video recordings the Employees made.

These records serve for Quality control and to Judgment whether the Investigator or the Therapists who prescribed treatments (exercises, Explanations , Intervention e n)ndeed lich carried out have . From each therapists become some Videotape randomly selected , to be assessed by trained employees. There is an assessment system for this purpose that is always focused only on the actions of the therapists. We ask for your permission to be allowed to record the therapy days and to evaluate the recording by trained employees who are sworn to secrecy (assessment of the therapists) to eat .

Possibility to Participation at further Investigations

Through the Participation in the Therapy study receive She the possibility two other voluntary ones Investigation to participate. Zurn a at one genetic study (possibly. Blood- or saliva sample, hair) , for others through c h Magnetic resonance imaging (fMRI). a brain imaging study .

Sequence the additional examinations

You should take part in both or one of the additional examinations If you want to take part, then Oberweisen we You to the with us close together working Working groups here at the University Clinic for Psychiatry and Psychotherapy (Prof. Dr. Alexander Rapp). You will be invited from there and will receive detailed information about the event upcoming investigation . For participation in these additional examinations, you will receive reimbursement of expenses or travel expenses. Note that you too You can withdraw your willingness to take part in these additional examinations at any time. Only after detailed explanation by the experts and the answer Alles yours Questions , become She asked , the therefor required Sign the declaration of consent .

To expecting Advantages through one participation at of the study

She are closely supervised and received free of charge a detailed one Diagnosis as well as free psychotherapeutic treatment treatment without long Waiting period. There are the best ones Chances that the course of bipolar disorder will be favorably influenced by reducing your frequency of depressive or manic episodes . these illness episodes are significantly milder. However, it can also happen that your Participation has no direct benefit
for your mental health. The overall results · this scientific study are intended to help ensure that fur the Group of people suffering from bipolar disorder effective treatment and relapse prevention can be found for suffering patients .

Possible Risks and Side effects

With psychotherapeutic treatment it can to a temporary one Stress caused by active engagement with discussed in the therapy sessions and possibly avoided so far Topics come. This can happen under certain circumstances to occur or increase from suicidal tendencies (thoughts or plans to commit suicide) occur. They are regularly visited Sui zidality questioned; In addition, if you have suicidal thoughts, we urge you to contact the study staff or appropriate (local) emergency services at any time

to turn around. Please inform us immediately of any deterioration in your health. independent of She a connection with the scientific study suspect.

Voluntariness the participation

Participation in this scientific study is voluntary. You can register your participation in the initial and follow-up examinations, the interviews, the magnetic resonance examination and stop the therapy offered at any time without giving reasons. This will not cause you any disadvantages. If you decide to withdraw from the therapy study early or your participation is terminated early for other reasons, it is in It is important to your own interests that you have a conversation with the director of studies drove to your Further treatment to to back up.

The scientific investigation will be terminated early if this is deemed necessary for ethical or medical reasons. If this unlikely event occurs, you will be informed about it and involved in the decision about possible further treatment included .

confidentiality the Oaten

The personal data and information from this scientific Study will be ver treated confidentially . Those within the framework of scientific The data collected in the study are pseudo nymized, i.e. without reference to or mention of your personal details, such as name, Address, in compliance with the legal data protection rules for employees Study after prior instruction and an express declaration of confidentiality, scientifically (statistically) evaluated. With consent to participate By participating in this scientific study you agree that She with the in the framework of this Recording carried out during the investigation van Oaten to yours health and the findings collected and their evaluation by scientific staff of the study . All persons who have access to your documents or evaluate the findings and scientific data are bound to data secrecy in accordance with the Federal Data Protection Act.
The data management of the entire study (with 9 clinical centers,) as well as the entire joint project "Bipolife" (with 7 large studies) is carried out via the Institute for Medical Informatics at the University Medical Center Gottingen. The oats are, in accordance with the data protection concept of the technology and method platform for networked medical research eV (TMF - Tele - Matic platform) only in pseudonymized form, separated from personal details such as name and birthday (say. IDAT) saved. The Administration the IDAT takes place Ober a trustworthy data trustee. This is for the Assignment between the identifying Oaten and the in the databases used Pseudonyms responsible.
The Data collection and -storage he follows by means of the professional Study software SecuTrial (www.sec u tr i al.com) . With SecuTrial can be multi-center Studies carried out become. SecuTrial

is GCP compliant (WHO Good Clinical Practice) and meets all international requirements Regulate for Therapy studies and Approvals (FDA Regulation 21 CFR part 11).
The database was carefully tested before use with regard to the questions, interviews and instruments used as well as the functions used. Only study directors and researchers have access to the data entered specially designated controllers (monitor). An export of the data for statistical purposes Evaluation becomes only to the one in Study protocol described points in time carried out and is reserved for the study director or the statistician of the study . The structure, processes and legal basis of data management are explained in detail in a separate data protection concept. You can already find this in numerous ethics committees and data protection officer checked and approved data protection concept at www . bi po l i f e . de . We are happy to provide you an expression of the data protection concept available.

The data collected as part of this therapy study will be stored pseudonymously (with no apparent connection to you, encrypted) for 10 years and will be used during this time for scientific, never personal evaluations. The respective users are scientists joint project, the before through the Study management and the Security board checked and the use of the oats has been approved.

Notice on Insurance
No separate insurance will be taken out for the planned examination, as if carried out properly he Handling the Investigations and the Psychotherapies for She no There is a recognizable and high risk. However, there is insurance cover through the university and clinic's usual liability insurance.

Status of this Information
This information takes into account all circumstances and facts that are known about SEKT and FEST as well as the examination procedures used here until June 2015. If there are any significant changes that should be brought to your attention as a participant in this scientific study, you will be informed immediately. If you have any further questions about the study, the project staff or I personally will be happy to assist you.

Prof. Dr. Martin Hautzinger

Supplementary Online Content

Hautzinger M; A2 BipoLife Consortium. Adjuvant psychotherapies to prevent relapse in bipolar disorder: a randomized clinical trial. *JAMA Psychiatry*. Published online June 5, 2024. doi:10.1001/jamapsychiatry.2024.1310

eAppendix 1. Adherence to Intervention Protocol

eAppendix 2. Social Functioning (Global Assessment of Functioning) Ratings

eAppendix 3. Kaplan-Meier Survival Curves for Time to First Relapse (New Episode) With Pointwise 95% Confidence Bands for Bipolar 1 and Bipolar 2, Independent of Psychotherapy Received

eAppendix 4. Kaplan-Meier Curves for Time to First Relapse (New Episode) for Bipolar 1 and Bipolar 2 Separated by Relapse Prevention Treatment (SEKT, FEST) Received

eAppendix 5. Proportional Hazards Model (Multivariate) Including 8 Covariates and Their Interaction

eAppendix 6. Propensitiy Score Matching and Estimates

eAppendix 7. Kaplan-Meier Curves for Time to First Relapse Demonstrating the Interaction Effect of Gender and Compliance (Adherence To Group Sessions)

eAppendix 8. Short Outline Of Treatment Protocols

eAppendix 9. Compliance With Medication Throughout Study Time

This supplementary material has been provided by the authors to give readers additional information about their work.

© 2024 American Medical Association. All rights reserved.

eAppendix 1: Adherence to Intervention Protocol

A rating scale of 24 items (10 FEST, 10 SEKT related, 4 general, nonspecific) was developed, based on previous trial experiences (Mayer&Hautzinger, 2012; Hautzinger&Welz, 2008; Schramm et al. 2017).

Raters

We employed three master-level psychology students to rate treatment adherence and competence. All of the raters were independent to the A2 study and received approximately 20 h of training in the use of the scale and introduction to SEKT and FEST. The focus of this training was to be able to identify the presence or absence as well as intensity of the specific SEKT and FEST behaviours in a videotaped therapy session.

Procedures

Overall, 117 videos of group treatment (n = 60 SEKT sessions, n = 57 FEST session) were assessed by the raters. In order to assess interrater reliability, thirty videos thereof were rated by all three raters. Videotapes of 60-90 minutes length were randomly selected (three videos per study therapist and therapy type). The raters were given the videos and were not informed which videotapes were of which therapy type. Raters watched the videos and completed the rating scale individually. To prevent rater drift all the raters met on occasion to review the ratings of single rated videos.

Inter-rater Reliability

The reliability of the ratings obtained from the three raters was examined by calculating intraclasscorrelation coefficients (ICC), treating the raters as random effects (Strout & Fleiss, 1979, Model 2). The reliability for both subscales SEKT and FEST was very high, $ICC_{(2,1)} = 0.98$, $p < .001$ for both subscales. The competence subscale demonstrated moderate reliability, $ICC_{(2,1)} = 0.70$, $p < .001$.

Internal Consistency

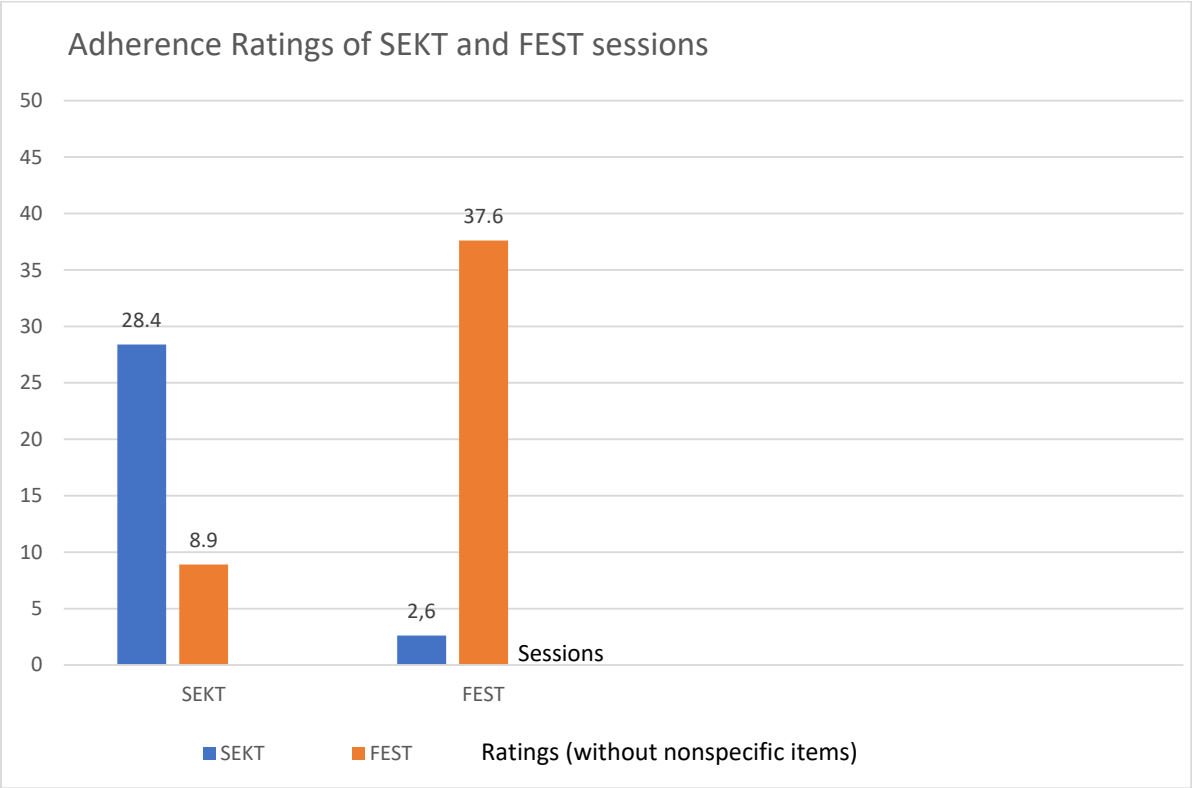
To determine whether the rating scale items were consistent with each other, we calculated Cronbachs α. The alpha for the SEKT and FEST subscale was 0.99 each, representing an excellent internal consistency. The competence subscale had a good internal consistency with $\alpha = .88$.

Treatment Differences (see figure)

To assess treatment adherence, we looked for treatment differences between the two therapy types by comparing the mean scores for each treatment group on the SEKT and FEST subscale. Adherence ratings of the SEKT treatment sessions were significantly higher on the

© 2024 American Medical Association. All rights reserved.

SEKT subscale (*M* = 28.44, *SD* = 6.30) than on the FEST subscale (*M* = 8.86, *SD* = 6.39), *t*(59) = 13.65, *p* = .001. In addition, adherence ratings of the SEKT subscale were significantly higher for SEKT treatment compared to FEST treatment (*M* = 3.53, *SD* =2.08), *t*(72.36) = -29.01, *p* <= .001. A similar pattern was observed for FEST, with significantly higher adherence ratings for the FEST treatment sessions on the FEST subscale (*M* = 37.57, *SD* =5.63) relative to the SEKT subscale (M = 2,6, SD 1,9), *t*(56) = 40.47, *p* = .001. Furthermore, adherence ratings of the FEST subscale were significantly higher for FEST treatment compared to SEKT treatment, *t*(114.38) = 25.83, *p* = .001.



Ratingscale to assess adherence to treatment protocol

1) Th. ist unterstützend, freundlich, optimistisch.	**0**	**1**	**2**	**3**	**4**	**5**
2) Th. modifiziert automatische Gedanken oder Grundüberzeugungen mit Hilfe von kognitiven Techniken.	**0**	**1**	**2**	**3**	**4**	**5**
3) Th. verzichtet auf die Verwendung jeglichen strukturförderlichen Materials.	**0**	**1**	**2**	**3**	**4**	**5**
4) Therapeutin/Therapeut hilft Patienten ihren Alltag/ihre Woche besser zu strukturieren und auf eine angemessene Balance von Tätigkeiten und Schlaf-Wach-Rhythmus zu achten.	**0**	**1**	**2**	**3**	**4**	**5**
5) Th. gelingt zutreffende Verbalisierungen emotionaler Erlebnisinhalte der Patienten.	**0**	**1**	**2**	**3**	**4**	**5**

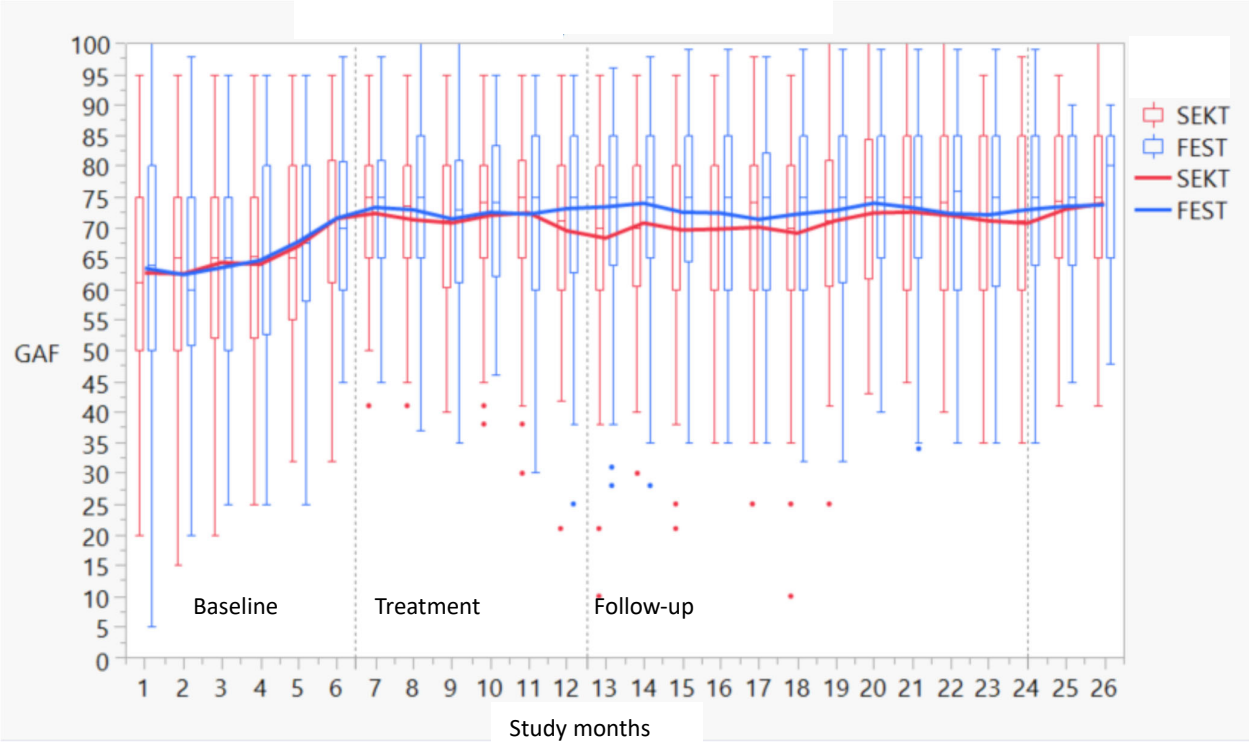
© 2024 American Medical Association. All rights reserved.

6)	Th. stellt Zusammenhänge zu Themen her, die für Patienten in einer vorherigen Sitzung wichtig gewesen waren bzw. zur Sprache kamen.	**0**	**1**	**2**	**3**	**4**	**5**
7)	Th. bietet in unsystematischer Weise Erklärungen, Informationen, an, ohne einem bestimmten Therapiemodell zu folgen und zu dominieren, verweist bestenfalls auf den Ratgeber.	**0**	**1**	**2**	**3**	**4**	**5**
8)	Th. bestärkt Patienten darin, ausreichende Ressourcen (Selbstkontrolle) zu besitzen, um den Verlauf ihrer Bipolaren Störung selbst beeinflussen zu können und Rückfälle zu verhindern.	**0**	**1**	**2**	**3**	**4**	**5**
9)	Th. spricht Patienten direkt an und bezieht Patienten in das Gruppengeschehen ein.	**0**	**1**	**2**	**3**	**4**	**5**
10)	Th. bestärkt (lobt) für Kooperation, Umsetzung oder Erprobung der Modulelemente.	**0**	**1**	**2**	**3**	**4**	**5**
11)	Th. geht während der Sitzung bei Äußerungen vor allem auf die Emotionen der Patienten ein.	**0**	**1**	**2**	**3**	**4**	**5**
12)	Th. betont die Bedeutung von Denken, Fühlen und Verhalten für die Stimmungsregulation, bietet kognitiv-behaviorale (Re)Formulierungen an.	**0**	**1**	**2**	**3**	**4**	**5**
13)	Th. spiegelt die Besorgnisse/Beschwerden der Patienten, um so das Besprechen dieser Sorgen zu erleichtern.	**0**	**1**	**2**	**3**	**4**	**5**
14)	Th. vermittelt den Eindruck eines interessierten, nicht-wertenden Zuhörers, der keinem festgelegten Plan folgt, sondern jedes von den Patienten initiierte Gespräch unterstützt.	**0**	**1**	**2**	**3**	**4**	**5**
15)	Th. bietet in geplanter Weise Erklärungen und Informationen (Psychoedukation) an, nimmt dabei Bezug auf das multifaktorielle (psychobiologische) Therapiemodell.	**0**	**1**	**2**	**3**	**4**	**5**
16)	Th. verhält sich geduldig, unterbricht nicht und gibt Patienten den benötigten Raum und die benötigte Zeit, egal wohin das führt.	**0**	**1**	**2**	**3**	**4**	**5**
17)	Th. stellt Zusammenhänge zu Themen früherer Module her (Überblick).	**0**	**1**	**2**	**3**	**4**	**5**
18)	Th. ermöglicht Patienten, Inhalt und Verlauf der Sitzung selbst zu bestimmen, gibt keine Struktur vor oder versucht eine einzuhalten.	**0**	**1**	**2**	**3**	**4**	**5**
19)	Th. benutzt die Therapie- und Informationsmaterialien, teilt diese an Teilnehmer aus.	**0**	**1**	**2**	**3**	**4**	**5**
20)	Th. ist aktiv und strukturiert, erstellt bzw. folgt dem Therapieplan und setzt die anstehenden Module, Übungen um.	**0**	**1**	**2**	**3**	**4**	**5**
21)	Th. unterstützt den Austausch von Erfahrungen zwischen den einzelnen Teilnehmern ohne ein Thema vorzugeben bzw. zu dominieren.	**0**	**1**	**2**	**3**	**4**	**5**
22)	Th. ermutigt Patienten, neue Verhaltensweisen und Aktivitäten anzuwenden und zu erproben.	**0**	**1**	**2**	**3**	**4**	**5**
23)	Th. macht einen sicheren, entspannten, ruhigen, kompetenten Eindruck.	**0**	**1**	**2**	**3**	**4**	**5**
24)	Th. bespricht Aufgaben, Übungen bzw. führt diese mit den Patienten durch.	**0**	**1**	**2**	**3**	**4**	**5**

© 2024 American Medical Association. All rights reserved.

eAppendix 2:

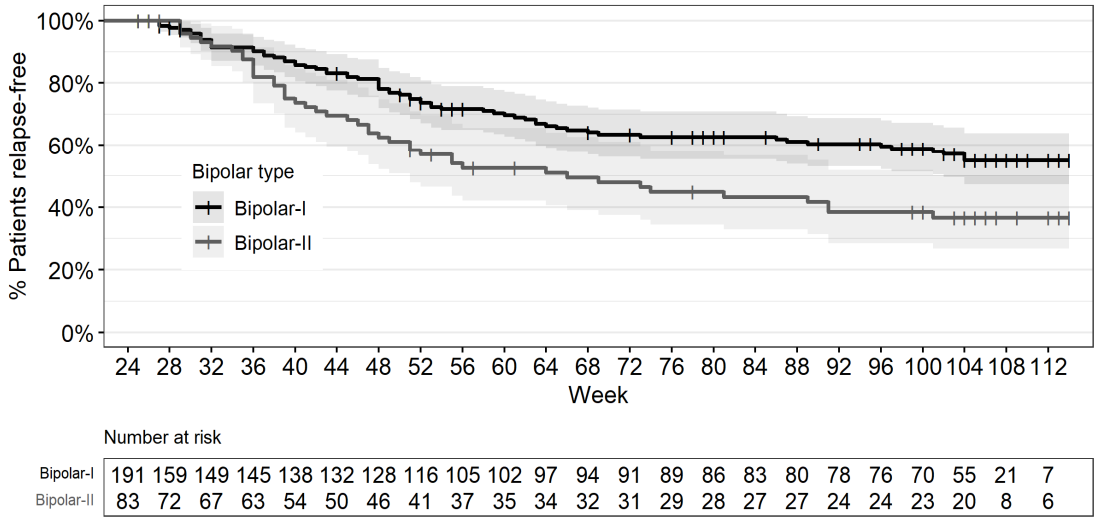
Social Functioning (Global Assessment of Functioning) ratings. Means and Box plots for total study time on a monthly basis.



© 2024 American Medical Association. All rights reserved.

eAppendix 3:

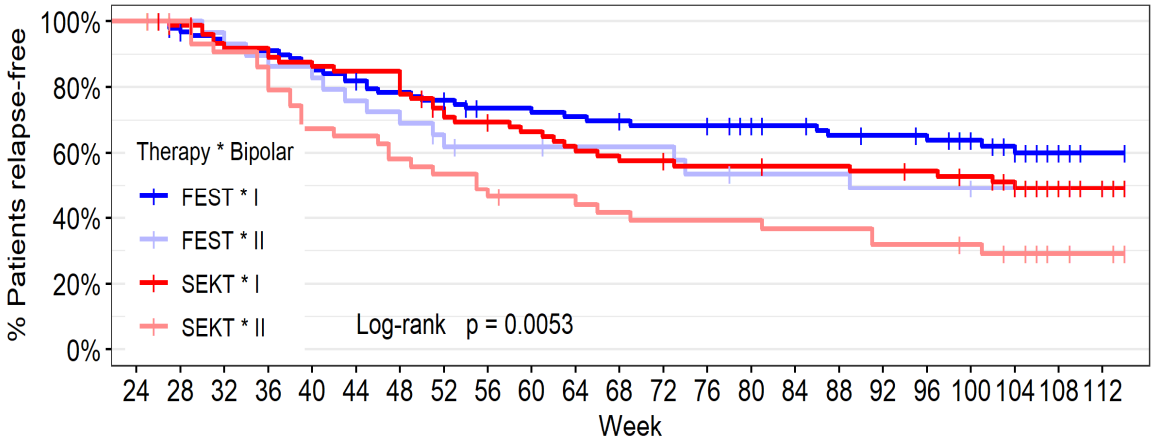
Kaplan-Meier survival curves for time to first relapse (new episode) with pointwise 95% confidence bands for Bipolar 1 and Bipolar 2, independent of psychotherapy received (N = 274). At the end of follow-up, 58% of Bipolar 1 group and 39% of Bipolar 2 group was relapse free.



© 2024 American Medical Association. All rights reserved.

eAppendix 4:

Kaplan-Meier curves for time to first relapse (new episode) for Bipolar 1 and Bipolar 2 separated by relapse prevention treatment (SEKT, FEST) received (N = 274). At the end of follow-up, 61% of Bipolar 1 treated by FEST, 49% of Bipolar 1 treated by SEKT, 49% of Bipolar 2 treated by FEST, and 30% of Bipolar 2 treated by SEKT were relapse free.



Number at risk

FEST * I	102	86	82	79	75	71	67	65	57	57	55	54	52	52	49	47	44	43	42	38	31	11	4
FEST * II	37	29	28	26	25	22	21	18	16	16	15	15	15	13	12	12	12	11	11	11	10	5	4
SEKT * I	89	73	67	66	63	61	61	51	48	45	42	40	39	37	37	36	36	35	34	32	24	10	3
SEKT * II	46	43	39	37	29	28	25	23	21	19	19	17	16	16	16	15	15	13	13	12	10	3	2

© 2024 American Medical Association. All rights reserved.

eAppenidx 5

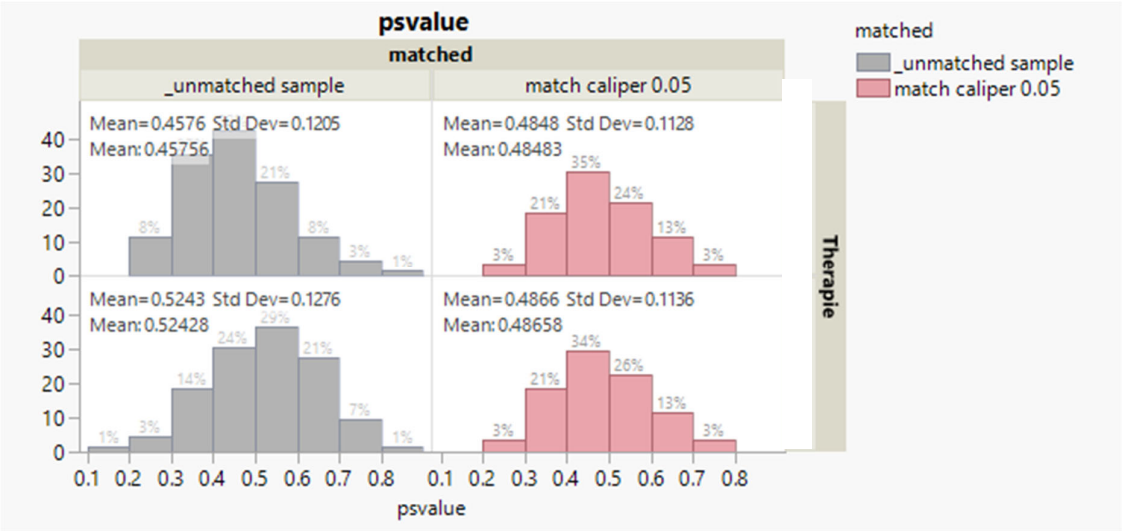
Proportional Hazards Modell (multivariate) including 8 covariates and their interaction (*).

Covariates and their interaction	p =
Gender*Compliance	0.00110 *
Bipolar 1/2	0.01624 *
Bipolar 1/2*previous psychotherapy	0.01699 *
Therapy*comorbidity	0.02227 *
Therapy*Bipolar 1/2	0.03118 *
Gender	0.05688
Age*comorbidity	0.06278
Bipolar 1/2*number of previous episodes	0.08966
Previous psychotherapy	0.11508
Number of previous episodes*previous psychotherapy	0.11692
Age*number of previous episodes	0.14095
Bipolar 1/2*comorbidity	0.14202
Compliance*comorbidity	0.19662
Compliance*previous psychotherapy	0.20849
Age*Compliance	0.22312
Compliance*number of previous episodes	0.22512
Age*previous psychotherapy	0.26569
Therapy*previous psychotherapy	0.28550
Compliance	0.28915
Comorbidity*previous psychotherapy	0.28967
Age*Bipolar 1/2	0.35953
Number of previous episodes	0.36911
Therapy*Gender	0.37627
Therapy*Compliance	0.39053
Therapy	0.58858
Gender*Age	0.75282
Compliance*Bipolar 1/2	0.77840
Comorbidity*number of previous episodes	0.85249
Gender*Bipolar 1/2	0.86136
Therapy*Age	0.86359
Comorbidity	0.86867
Age	0.89068
Gender*previous psychotherapy	0.90670
Gender*number of previous episodes	0.95049
Gender*Comorbidity	0.95760
Therapy*number of previous episodes	0.98583

© 2024 American Medical Association. All rights reserved.

eAppendix 6

Propensitiy Score Matching



Parameter estimates

Term	Estimate	Std Error	Lower 95%	Upper 95%
Study site 1	-0.193884	0.3768109	-0.93242	0.5446518
Study site 2	-1.365557	0.5625459	-2.468127	-0.262987
Study site 3	-0.006844	0.3250780	-0.643985	0.630297
Study site 4	-0.325014	0.3615172	-1.033574	0.3835468
Study site 5	-0.574581	0.6120239	-1.774125	0.6249643
Study site 6	1.3293489	0.5657354	0.2205279	2.4381698
Study site 7	0.6394053	0.7990317	-0.926668	2.2054786
Study site 8]	0.2103207	0.5486255	-0.864966	1.285607
Therapy	-0.1141334	0.2399293	-0.584386	0.3561194
Gender	0.2915809	0.2481293	-0.194744	0.7779054
Age	0.1635785	0.2551006	-0.33641	0.6635666
Compliance	0.3273945	0.2871898	-0.235487	0.8902762
Bipolar I/II	-0.2081976	0.2786789	-0.754398	0.338003
Comorbidity	-0.2277053	0.2476504	-0.713091	0.2576805
Previous episodes	0.3580506	0.3495479	-0.327051	1.0431519
Previous psychotherapy	0.7696138	0.3365959	0.109898	1.4293297
Therapy * Gender	0.1273762	0.1613770	-0.188917	0.4436692
Therapy * Age	0.3283051	0.1619892	0.0108121	0.645798
Therapy * Compliance	-0.0883620	0.2044963	-0.489167	0.3124434
Therapy * Bipolar I/II	0.3758564	0.1763967	0.0301252	0.7215876
Therapy * Comorbitiy	0.4450139	0.1789425	0.094293	0.7957349
Therapy * Previous episodes	-0.2002548	0.1777239	-0.548587	0.1480776
Therapy * Previous psychotherapy	0.0489779	0.1873613	-0.318243	0.4161992
Gender * Age	-0.0607662	0.1749758	-0.403713	0.2821801
Gender * Compliance	0.5575680	0.1913129	0.1826016	0.9325344
Gender * Bipolar I/II	-0.1561235	0.1700099	-0.489337	0.1770899
Gender * Comorbidity	-0.1322040	0.1845342	-0.493884	0.2294764
Gender * Previous episodes	-0.0430684	0.1776709	-0.391297	0.3051601
Gender * Previous psychotherapy	-0.0636506	0.1982668	-0.452246	0.3249451
Age * Compliance	-0.4657330	0.2289675	-0.914501	-0.016965
Age * Bipolar I/II	-0.3254421	0.1903071	-0.698437	0.047553
Age * Comorbidity	0.1682858	0.1879609	-0.200111	0.5366824

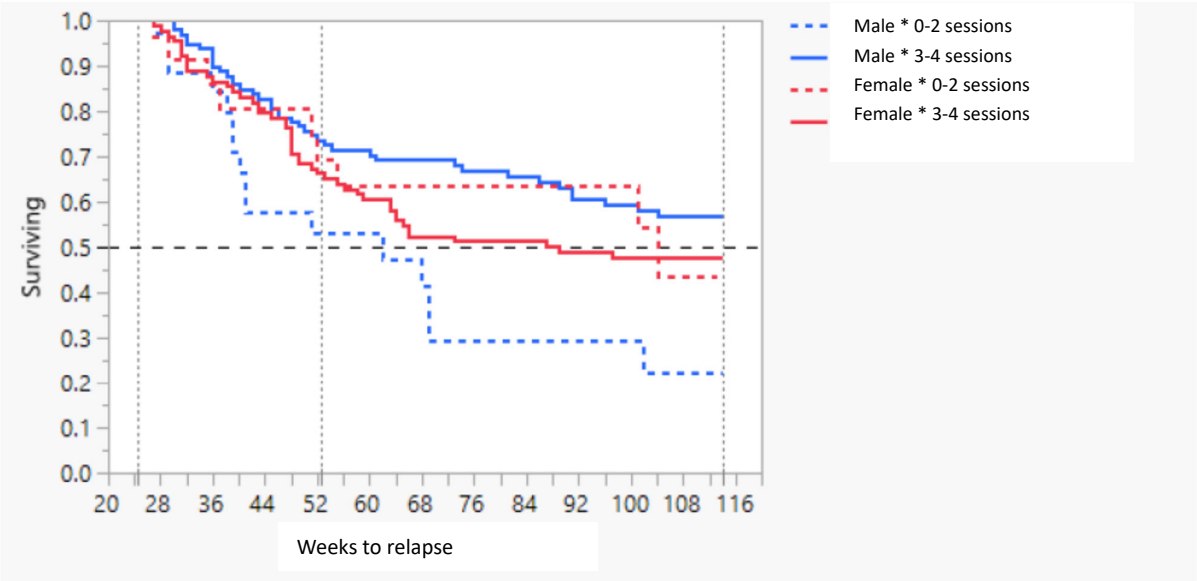
© 2024 American Medical Association. All rights reserved.

Term	Estimate	Std Error	Lower 95%	Upper 95%
Age * Previous episodes	0.1738774	0.1805330	-0.179961	0.5277156
Age * Previous psychotherapy	-0.3968673	0.2018304	-0.792448	-0.001287
Compliance * Bipolar I/II	0.0415889	0.2126872	-0.37527	0.4584482
Compliance * Comorbidity	-0.2100108	0.2294838	-0.659791	0.2397691
Compliance * Previous episodes	0.6165833	0.2697000	0.087981	1.1451855
Compliance * Previous psychotherapy	0.5173692	0.2740309	-0.019722	1.0544598
Bipolar I/II * Comorbidity	0.2343097	0.1988751	-0.155478	0.6240977
Bipolar I/II * Previous episodes	0.4409532	0.2054018	0.0383731	0.8435333
Bipolar I/II * Previous psychotherapy	0.4802020	0.2254296	0.0383681	0.9220359
Comorbidity * Previous episodes	0.1828288	0.2115903	-0.231881	0.5975383
Comorbiditiy * Previous psychotherapy	0.2177806	0.2466394	-0.265624	0.7011849
Previous episodes * Previous psychotherapy	-0.2426480	0.2566142	-0.745603	0.2603066

© 2024 American Medical Association. All rights reserved.

eAppendix 7

Kaplan-Meier curves for time to first relapse demonstrating the interaction effect of Gender and Compliance (adherence to group sessions). Men missed more sessions than women resulting in poor outcome



© 2024 American Medical Association. All rights reserved.

eAppendix 8

Short outline of treatment protocols

SEKT

The basic or general modul applied mindfulness training and self-observation of daily mood, activities, social contacts, sleep behavior, drug taking during the first 90 minutes of each treatment day.

Modul 1 focuses on psychoeducation about Bipolar disorder, its development and course of illness by using life-charts, understanding the association of stress, risk factors, behavior, depressive and (hypo-) manic episodes, finding life balance and personal sleep-wake (work, social) rhythm.

Modul 2 teaches early warning signs of mood changes, personal risk factors and how to control them, stress management, problem solving and interpersonal skills.

Modul 3 works on cognitive processes, ruminations, dysfunctional ways of thinking, changing belief systems and schemata.

Modul 4 helps to understand emotional processes and impulses, develope tolerance and acceptance towards thoughts, techniques to distract and regulate strong emotions. Alle moduls have materials, work sheets, exercises, and home work. Depending on the group of study participants and their needs, therapist can select certain topics and exercises.

FEST

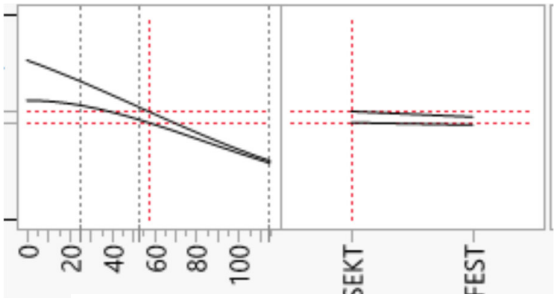
Participants are encouraged to come up with their personal experience, acute or chronic problems, feelings, concerns and life topics. Participants interact with each other, ask and make comments, report their own experiences, their perceptions and interpretations. Therapists do not structure the ongoing interaction between group members or intervene by coming up with exercises, problem solving strategies, or homework. They focus on emotions, verbalize their observations of emotions behind a certain story, are empathetic, positive, let the group go their own way.

© 2024 American Medical Association. All rights reserved.

eAppendix 9

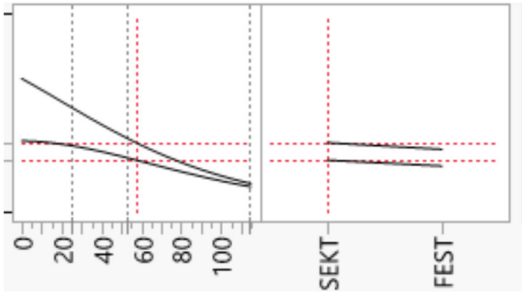
Compliance with Medication throughout study. The following figure are based on information given by participants (unclear validity). There is never a relevant difference between treatments or BD 1 and BD 2 (Chi squared 4.46442 and 3.9471, $p < 0.11$). Because of a great number of missing information which is increasing over time (see left side of figures below), we did not do more statistics.

Mood Stabilizer



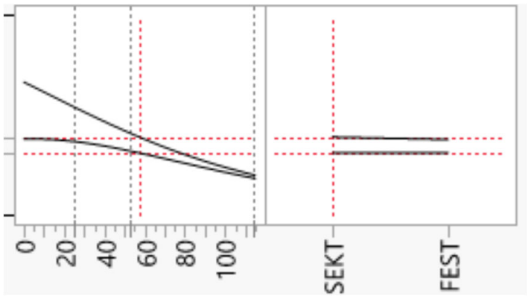
Time Intervention over time

Antidepressants



Time Intervention over time

Neuroleptics



Time Intervention over time

© 2024 American Medical Association. All rights reserved.

Data Sharing Statement

Hautzinger. Adjuvant Psychotherapies to Prevent Relapse in Bipolar Disorder. *JAMA Psychiatry*. Published June 05, 2024. doi:10.1001/jamapsychiatry.2024.1310

Data
Data available: Yes
Data types: Deidentified participant data
How to access data: hautzinger@uni-tuebingen.de
When available: With publication

Supporting Documents
Document types: None

Additional Information
Who can access the data: researchers after approval of planned use
Types of analyses: for any purpose
Mechanisms of data availability: with investigator support, after approval of a proposal